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Sheaf cohomology

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Introduction

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PART I

Category theory

Introduction

Why categories? Historical introduction?

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category). A **category** C consists of:

- a class $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection of sets $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then

$$(f \circ g) \circ h = f \circ (g \circ h)$$

- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then

$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

Definition 1.1.2 (Small category). A category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set.

Remark 1.1.1 (Size issues). We will be sketchy about foundational issues (set-theoretic issues, universes, ...).
Therefore all categories will be $\mathcal{U}$-categories for a large enough *universe* $\mathcal{U}$ (i.e: for any pair of objects $X, Y \in \mathsf{C}$ the Hom-set $\mathrm{Hom}(X, Y)$ is $\mathcal{U}$-small) and small categories will be $\mathcal{U}$-small (the set of objects $\mathrm{Ob}(\mathsf{C})$ is $\mathcal{U}$-small).
We will refer as "big" categories those that are $\mathcal{V}$-categories for a larger universe $\mathcal{V}$. By Grothendieck's axiom (for any set X there exists a universe $\mathcal{U}$ such that $X \in \mathcal{U}$) every category C is a $\mathcal{V}$-category for a big enough universe $\mathcal{V}$, and this universe can be chosen so that C is $\mathcal{V}$-small.
For a basic introduction of what an *universe* is and a more carefull treatment of this type of size issues see [KS06, Section 1.1]. More details can also be found at [SGA4]

Notation. To indicate that X is an object of a category C we write

$$X \in \mathsf{C} \text{ instead of } X \in \mathrm{Ob}(\mathsf{C})$$

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphism from X to Y we write

$$f : X \to Y \text{ or } X \xrightarrow{f} Y$$

We will sometimes abbreviate the composition $f \circ g$ as fg.

Example 1.1.1.

- The category Set has the collection of all sets as objects and mappings as morphisms.
- The category Grp has groups as objects and group homomorphisms as morphisms.
- The category Ab has abelian groups as objects and group homomorphisms as morphisms. It is a full subcategory of Grp.
- The category Ring has rings as objects and ring homomorphisms as morphisms.[a]
- The category Rng has (not necessarilly unital) rings as objects and ring homomorphisms as morphism.
- The category Top has topological spaces as objects and continuous functions as morphisms.
- For a fixed ring R, the category $\mathsf{Mod}(R)$ has (left) R-modules as objects and R-linear maps as morphisms.
- For a fixed field k, the category Vect_k has k-vector spaces as objects and k-linear maps as morphisms.

[a] We adopt the convetion that all rings have an unit and all ring homomorphisms preserve it.

Examples of non-concrete categories?

Example 1.1.2 (Discrete category)**.** Given a category C we can define the **discrete category** C_d taking the same collection of objects as C but only the identity morphisms between them.

Definition 1.1.3 (Subcategory)**.** Given a category C, a **subcategory** D consists of a subcollection of objects and a subcollection of morphisms in C such that

- If $f : X \to Y$ is a morphism in D then X, Y are objects in D.
- If $f : X \to Y$ and $g : Y \to Z$ are morphisms in D, the composition $gf : X \to Z$ is in D.
- If X is an object in D then the identity map $\mathbb{1}_X$ is in D.

We say that D is a **full subcategory** if for all objects X, Y in D every morphism $f : X \to Y$ in C is also in D.

Example 1.1.3.

- Ab is a full subcategory of Grp.
- The category of finite sets FinSet is a full subcategory of Set.
- Ring is a subcategory of Rng that is not full.
- If k is a field, Vect_k is a full subcategory of $\mathsf{Mod}(k)$.

Definition 1.1.4 (Product category)**.** Let C, D be two categories. The **product category** $\mathsf{C} \times \mathsf{D}$ is the category consisting of

- As objects: ordered pairs (C, D), where C is an object in C and D is an object in D, i.e:
$$\mathrm{Ob}(\mathsf{C} \times \mathsf{D}) := \mathrm{Ob}(\mathsf{C}) \times \mathrm{Ob}(\mathsf{D})$$
- As morphisms: ordered pairs $(C \xrightarrow{f} C', D \xrightarrow{g} D')$, where $C \xrightarrow{f} C'$ is a morphism in C and $D \xrightarrow{g} D'$ is a morphism in D, i.e: for all objects $(C, D), (C', D')$ in $\mathsf{C} \times \mathsf{D}$
$$\mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D')) := \mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D')$$
- Composition is defined *componentwise*: if $(C \xrightarrow{f} C', D \xrightarrow{g} D')$ and $(C' \xrightarrow{f'} C'', D' \xrightarrow{g'} D'')$ are two morphisms in $\mathsf{C} \times \mathsf{D}$ then
$$(f', g') \circ (f, g) := (f' \circ f, g' \circ g) : C \times D \to C'' \times D''$$

Example 1.1.4 (Category of open sets)**.** Let X be a topological space.
The category $\mathsf{Open}(X)$ consisting of

- As objects, the open subsets of X.
- As morphisms, inclusion of open sets: $U \to V$ if and only if $U \subseteq V$

This is a small category.

1.1.1 Duality

Definition 1.1.5 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.
- If $f^{\mathrm{op}} \colon X \to Y$, $g^{\mathrm{op}} \colon Y \to Z$ are morphisms in C^{op} their composition is defined by
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$$
where $f \colon Y \to X$ and $g \colon Z \to Y$ are the associated morphisms in C.

Example 1.1.5. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.
The opposite category P^{op} corresponds to the same set P equipped with the reversed ordering $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P.

Remark 1.1.2. Let $f^{\mathrm{op}} \colon Y \to X$ and $g^{\mathrm{op}} \colon Y \to X$ be two morphisms in the opposite category C^{op}. Then $f^{\mathrm{op}} = g^{\mathrm{op}}$ in C^{op} if and only if $f = g$ in C.

A categorical statement P^* is defined as the **dual** of a statement P in a category C if P^* holds in the opposite category C^{op} if and only if P holds in C.

Note that, by definition, for any category C we have that $(\mathsf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathsf{C}$.

Therefore if a statement is true for all categories C, its dual is true for all categories C^{op}. Since C^{op} is an arbitrary category the dual statement is also true for all categories.

In practice, this means that proving an statement directly proves its dual for free.

1.2 Types of morphisms

Definition 1.2.1 (Monomorphism and epimorphism). Let $f : X \to Y$ be a morphism in a category C. We say that f is a

- **Monomorphism** (or is **monic**) if for every pair of parallel morphisms $g, h : Z \to X$ in C, the equality $f \circ g = f \circ h$ implies $g = h$.
- **Epimorphism** (or is **epic**) if for every pair of parallel morphisms $g, h : Y \to Z$ in C, the equality $g \circ f = h \circ f$ implies $g = h$.

Example 1.2.1.

- In Set, an monomorphism (resp. epimorphism) is an injective (resp. surjective) function.
- In Grp, an monomorphism (resp. epimorphism) is an injective (resp. surjective) group homomorphism.
- In Top, an monomorphism (resp. epimorphism) is an injective (resp. surjective) continuous map.

Lemma 1.2.1 (Monomorphisms and epimorphisms are dual). Let $f : X \to Y$ be a morphism in a category C. Consider the associated morphism $f^{\mathrm{op}} : Y \to X$ in the opposite category C^{op}.
Then f is a monomorphism (resp. epimorphism) in C if and only if f^{op} is an epimorphism (resp. epimorphism) in C^{op}.

Proof. Consider of pair of morphisms $g, h : Z \rightrightarrows X$ in C and their opposites $g^{\mathrm{op}}, h^{\mathrm{op}} : X \rightrightarrows Z$ in C^{op}.

Note that, by definition of composition in C^{op} we have

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}}$$

Therefore we have that $f \circ g = f \circ h$ implies $g = h$ if and only if $(f \circ g)^{\mathrm{op}} = (f \circ h)^{\mathrm{op}}$ implies $g = h$ if and only if $g^{\mathrm{op}} \circ f^{\mathrm{op}} = h^{\mathrm{op}} \circ f^{\mathrm{op}}$ implies $g^{\mathrm{op}} = h^{\mathrm{op}}$. □

Definition 1.2.2 (Isomorphism). A morphism $f : X \to Y$ in a category C is an **isomorphism** if there exists another morphism $g : Y \to X$ in C such that

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f and is denoted by f^{-1}.

Proposition 1.2.1. Let $f : X \to Y$ and $g : Y \to Z$ be isomorphisms in a category C. Then:

1. The inverse f^{-1} of f is unique.
2. The identity $\mathbb{1}_X$ is an isomorphism for every $X \in \mathsf{C}$.
3. $g \circ f$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. If $g \circ f$ is an isomorphism and f is an isomorphism, then g is an isomorphism. Analogously if $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Proof.

1. Suppose that $h_1, h_2 : Y \to X$ are inverses of f. We have
$$h_1 = h_1 \circ \mathbb{1}_Y = h_1 \circ (f \circ h_2) = (h_1 \circ f) \circ h_2 = \mathbb{1}_X \circ h_2 = h_2$$
2. For all $X \in \mathsf{C}$ by definition
$$\mathbb{1}_X \circ \mathbb{1}_X = \mathbb{1}_X$$
i.e: $\mathbb{1}_X$ is its own left and right inverse.
3. We only need to check that $f^{-1} \circ g^{-1}$ is both a left and right inverse of $g \circ f$. On the one hand
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathbb{1}_X$$
On the other
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathbb{1}_Z$$
4. Let $h \coloneqq (g \circ f)^{-1} : Z \to X$. We are going to check that $f \circ h : Z \to Y$ is the inverse of g. On the one hand
$$g \circ (f \circ h) = (g \circ f) \circ h = \mathbb{1}_Z$$
On the other, since $h \circ (g \circ f) = \mathbb{1}_X$ precomposing with f and postcomposing with f^{-1} we have
$$\begin{aligned} f \circ (h \circ g \circ f) \circ f^{-1} = f \circ \mathbb{1}_X \circ f^{-1} = \mathbb{1}_Y &\implies (f \circ h) \circ g \circ (f \circ f^{-1}) = \mathbb{1}_Y \\ &\implies (f \circ h) \circ g = \mathbb{1}_Y \end{aligned}$$

□

Example 1.2.2.

- In Set, an isomorphism is a bijection.
- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

Proposition 1.2.2. Let $f : X \to Y$ be an isomorphism. Then f is both a monomorphism and epimorphism.

Proof. Suppose $f : X \to Y$ is an isomorphism in a category C. Then there exists a morphism $f^{-1} : Y \to X$ in C such that
$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$
Consider a pair of morphisms $g, h : Z \rightrightarrows X$ such that $f \circ g = f \circ h$. Precomposing with f^{-1} we get
$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \implies (f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h \implies g = h$$
Therefore f is a monomorphism.

Analogously, if we consider pair of morphism $g', h' : Y \rightrightarrows Z$ such that $g' \circ f = h' \circ f$ this time by postcomposing with f^{-1} we get $g' = h'$, i.e: f is a epimorphism. □

Example 1.2.3. In general, a morphism $f: X \to Y$ that is both a monomorphism and epimorphism is not an isomorphism.
Indeed, in the category Top of topological spaces consider a set with more than one point X endowed with the discrete X_d and trivial X_t topology.
Consider the identity map $\mathbb{1}_X: X_d \to X_t$, which is continuous. Since $\mathbb{1}_X$ is injective we have that it is a monomorphism and, since it is surjective, it is an epimorphism. However since there does not exists a continuous bijection $X_t \to X_d$, $\mathbb{1}_X$ cannot be an isomorphism.

Proposition 1.2.3 (Llevar?). Let D be a subcategory of a category C. If $f: X \to Y$ is an isomorphism in D then f is also an isomorphism in C.

Proof. Since $f: X \to Y$ is an isomorphism in D there exists an inverse $f^{-1}: Y \to X$ in D. Therefore since $\mathrm{Hom}_{\mathsf{D}}(Y, X) \subseteq \mathrm{Hom}_{\mathsf{C}}(Y, X)$, f^{-1} is an inverse of f in C. □

Proposition 1.2.4 (Llevar?). Let D be a full subcategory of a category C. If $f: X \to Y$ is an isomorphism in C and $X, Y \in \mathsf{D}$ then f is also an isomorphism in D.

Proof. Since $f: X \to Y$ is an isomorphism in C there exists an inverse $f^{-1}: Y \to X$ in D. Since D is a full subcategory of D and $X, Y \in \mathsf{D}$ we have that $f^{-1} \in \mathrm{Hom}_{\mathsf{D}}(Y, X)$. Therefore f admits an inverse in D. □

1.3 Special objects

Definition 1.3.1 (Initial and terminal objects). An object $I \in \mathsf{C}$ is **initial** if for every object $X \in \mathsf{C}$ there is a unique morphism $I \to X$.
Dually, an object $T \in \mathsf{C}$ is **terminal** if for every object $X \in \mathsf{C}$ there is a unique morphism $X \to T$.

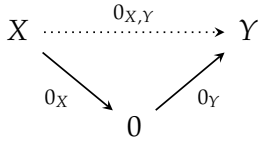
Proposition 1.3.1 (Uniqueness of initial and terminal objects). Let X, Y be two initial (resp. terminal) objects in a category C. Then there exists a unique isomorphism $X \xrightarrow{\sim} Y$ in C.

Proof. Since X is initial in C there exists a unique morphism $f: X \to Y$. Analogously there exist a unique morphism $g: Y \to X$.

Note that $g \circ f: X \to X$. Therefore since X is initial $g \circ f = \mathbb{1}_X$. Analogously since $f \circ g: Y \to Y$ we have $f \circ g = \mathbb{1}_Y$.

Therefore f and g are inverses of each other, i.e: $X \cong Y$. Moreover they are unique by definition. □

Definition 1.3.2 (Zero object). An object $0 \in \mathsf{C}$ is a **zero** object if it is both initial and terminal.
Note that for any two objects $X, Y \in \mathsf{C}$ there exist unique maps $0_X: X \to 0$ and $0_Y: 0 \to Y$. We call this composite map the **zero map** between X and Y and denote it $X \xrightarrow{0_{X,Y}} Y$:



By abuse of notation, we will denote all of these maps by 0.

Corollary 1.3.1 (Uniqueness of zero objects)**.** Let $0, \widetilde{0}$ be two zero objects in a category C. Then there exists a unique isomorphism $0 \xrightarrow{\sim} \widetilde{0}$ in C.

Proof. Since 0 and $\widetilde{0}$ are in particular initial we can apply Proposition 1.3.1. □

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor). Let C, D be two categories. A (covariant) **functor** $F : \mathsf{C} \to \mathsf{D}$ consists of:

- for each object $X \in \mathsf{C}$, an object $F(X) \in \mathsf{D}$.
- for each morphism $f : X \to Y$ in C, a morphism $Ff : F(X) \to F(Y)$ in D,

such that:

- $F(\mathbb{1}_X) = \mathbb{1}_{F(X)}$ for every $X \in \mathsf{C}$.
- $F(g \circ f) = Fg \circ Ff$ for every composable pair of morphisms f, g in C.

A functor is **contravariant** if it reverses the order of composition: $F(g \circ f) = Ff \circ Fg$. This is equivalent to a covariant functor $F : \mathsf{C}^{\mathrm{op}} \to \mathsf{D}$.
Diagramatically, a covariant functor is represent by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

And a contravariant functor by

$$\begin{array}{ccc} F\colon \mathsf{C} & \longrightarrow & \mathsf{D} \\ X & & F(X) \\ {\scriptstyle f}\downarrow & \longmapsto & \uparrow{\scriptstyle Ff} \\ Y & & F(Y) \end{array}$$

Remark 2.1.1. We will disfavour the covariant/contravariant distinction. Therefore all functors will be covariant (unless otherwise specified) and contravariant functors will be covariant functors from the opposite category.

Example 2.1.1 (Identity functor). Let C be a category. There exists an identity functor

$$\begin{array}{ccc} \mathbb{1}\colon \mathsf{C} & \longrightarrow & \mathsf{C} \\ X & & X \\ {\scriptstyle f}\downarrow & \longmapsto & \downarrow{\scriptstyle f} \\ Y & & Y \end{array}$$

Example 2.1.2 (Constant functor). Let C be a category. For every object $C \in \mathsf{C}$ there exists a constant functor

$$\begin{array}{rccc} \mathrm{ct}_C\colon & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & C \\ & f\downarrow & \longmapsto & \downarrow \mathbb{1}_C \\ & Y & & C \end{array}$$

Example 2.1.3 (Bifunctor). Let C, C' and D be three categories. Recall that we can construct the product category $\mathsf{C} \times \mathsf{C}'$.
A **bifunctor** is a functor of the form $F\colon \mathsf{C} \times \mathsf{C}' \to \mathsf{D}$. This means that for any $X \in \mathsf{C}$ and $X' \in \mathsf{C}$, $F(X,-)\colon \mathsf{C}' \to \mathsf{D}$ and $F(-,X')\colon \mathsf{C} \to \mathsf{D}$ are functors.
Furthermore for any morphisms $f\colon X \to Y$ in C and $g\colon X' \to Y'$ in C' we have a commutative diagram

$$\begin{array}{ccc} F(X,X') & \xrightarrow{F(X,g)} & F(X,Y') \\ {\scriptstyle F(f,X')}\downarrow & \searrow^{F(f,g)} & \downarrow{\scriptstyle F(f,Y')} \\ F(Y,X') & \xrightarrow[F(Y,g)]{} & F(Y,Y') \end{array}$$

Example 2.1.4. Let C be a For every object $C \in \mathsf{C}$ there exists a *covariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(C,-)\colon & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \downarrow f\circ \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

and a *contravariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(-,C)\colon & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f\downarrow & \longmapsto & \uparrow \circ f \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

We can combine this two into a bifunctor

$$\begin{array}{rccc} \mathrm{Hom}(-,-)\colon & \mathsf{C}^{\mathrm{op}} \times \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & (X,Y) & & \mathrm{Hom}(X,Y) \\ & (f,g)\downarrow & \longmapsto & \downarrow g\circ - \circ f \\ & (Z,W) & & \mathrm{Hom}(Z,W) \end{array}$$

Example 2.1.5. Let D be a subcategory of a category C. There exist an inclussion functor

$$\begin{array}{rccc} \iota\colon & \mathsf{D} & \hookrightarrow & \mathsf{C} \\ & X & & X \\ & f\downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.6 (Preimage functor). Let X, Y be two topological spaces. Consider their categories of open sets $\mathsf{Open}(X), \mathsf{Open}(Y)$.

For every continuous map $f : X \to Y$ we have a preimage functor

$$\begin{array}{ccc} f^{-1}\colon \mathsf{Open}(Y) & \longrightarrow & \mathsf{Open}(X) \\ U & & f^{-1}(U) \\ \downarrow & \longmapsto & \downarrow \\ V & & f^{-1}(V) \end{array}$$

Proposition 2.1.1 (Functors preseve isomorphisms)**.** Let C, D be two categories and $F\colon \mathsf{C} \to \mathsf{D}$ a functor between them. If f is an isomorphism in C then Ff is an isomorphism in D.

Proof. Let $f : X \to Y$ be a isomorphism in C. Then there exists an inverse $f^{-1}\colon Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Applying F to this equalities we have

$$\begin{aligned} F(f^{-1} \circ f) = F\mathbb{1}_X &\implies Ff^{-1} \circ Ff = \mathbb{1}_{F(X)} \\ F(f \circ f^{-1}) = F\mathbb{1}_Y &\implies Ff \circ Ff^{-1} = \mathbb{1}_{F(Y)} \end{aligned}$$

i.e: Ff has an inverse Ff^{-1} in D. □

Definition 2.1.2 (Functor types)**.** A functor $F\colon \mathsf{C} \to \mathsf{D}$ is:

- **faithful** if for every pair of objects $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is injective.
- **full** if for every pair $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is surjective.
- **fully faithful** if it is both full and faithful.
- **essentially surjective** if for every object $Y \in \mathsf{D}$ there exists an object $X \in \mathsf{C}$ such that $F(X) \cong Y$.

Example 2.1.7. A subcategory D of a category C is full if and only if the inclussion functor $\iota : \mathsf{D} \to \mathsf{C}$ is full.

2.2 Natural transformations

Definition 2.2.1. Let $F, G\colon \mathsf{C} \to \mathsf{D}$ be two functors. A **natural transformation** $\psi\colon F \implies G$ consists of a family of morphisms $\psi_X : F(X) \to G(X)$ in D, one for each object $X \in \mathsf{C}$, called the **components** of ψ, such that for every morphism $f : X \to Y$ in C, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\psi_X} & G(X) \\ {\scriptstyle Ff}\downarrow & & \downarrow{\scriptstyle Gf} \\ F(Y) & \xrightarrow[\psi_Y]{} & G(Y) \end{array}$$

Diagramatically, a natural transformations is represented by

$$\mathsf{C} \;\overset{F}{\underset{G}{\Downarrow \psi}}\; \mathsf{D}$$

If each component ψ_X is an isomorphism in D, we say that ψ is a **natural**

isomorphism, and we write $F \cong G$.

Definition 2.2.2 (Equivalent categories)**.** Two categories C, D are **equivalent** if there exist functors $F : \mathsf{C} \to \mathsf{D}$ and $G : \mathsf{D} \to \mathsf{C}$ such that $G \circ F \cong \mathbb{1}_{\mathsf{C}}$ and $F \circ G \cong \mathbb{1}_{\mathsf{D}}$. In this case, F (and G) is called an **equivalence of categories**.

Theorem 2.2.1 ([Rie, Theorem 1.5.9])**.** A functor $F : \mathsf{C} \to \mathsf{D}$ defining an equivalence of categories is fully faithful and essentially surjective. Assuming the axiom of choice, any functor with these properties defines an equivalence of categories.

Chapter 3

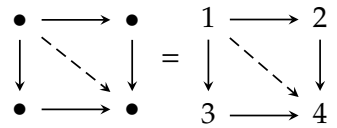
Limits and colimits

3.1 Diagrams, cones and cocones

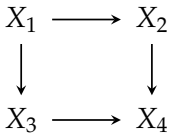
Definition 3.1.1 (Diagram). Let J be a small category.
A **diagram** X in a category C with shape J is a functor $X\colon \mathsf{J} \to \mathsf{C}$. For every $j \in \mathsf{J}$ we write $X_j \coloneqq X(j)$.

Example 3.1.1 (Constant diagram). A diagram $X\colon \mathsf{J} \to \mathsf{C}$ is **constant** if it is a constant functor const_c.
In this case the data of the diagram corresponds to the data given by a single object $c \in \mathsf{C}$.

Example 3.1.2. Consider the category

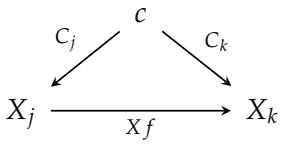
$\mathsf{J} =$  $=$

A diagram with shape J is a commutative square



3.1.1 Cones and cocones

Definition 3.1.2 (Cone). A **cone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a natural transformation $C\colon \mathrm{const}_c \Rightarrow X$ from the constant diagram to X. The components $\{C_j\colon c \to X_j\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *apex* c is a collection of morphisms $C_j\colon c \to X_j$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes



If $C\colon \mathrm{const}_c \Rightarrow X$ and $D\colon \mathrm{const}_d \Rightarrow X$ are two cones over X a morphism of cones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram

commute

$$\begin{array}{ccc} \mathrm{const}_c & \Longrightarrow & \mathrm{const}_d \\ & {\scriptstyle C}\searrow \quad \swarrow{\scriptstyle D} & \\ & X & \end{array}$$

Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cones $\Delta \downarrow X$.

Definition 3.1.3 (Cocone)**.** A **cocone** C over a diagram $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a natural transformation $C\colon X \Rightarrow \mathrm{const}_c$ from the constant diagram to X. The components $\{C_j\colon X_j \to c\}_{j\in\mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cocone C over $X\colon \mathsf{J} \to \mathsf{C}$ with *nadir* c is a collection of morphisms $C_j\colon X_j \to c$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes

$$\begin{array}{ccc} X_j & \xrightarrow{Xf} & X_k \\ & {\scriptstyle C_j}\searrow \quad \swarrow{\scriptstyle C_k} & \\ & c & \end{array}$$

If $C\colon X \Rightarrow \mathrm{const}_c$ and $D\colon X \Rightarrow \mathrm{const}_d$ are two cocones over X a morphism of cocones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram commute

$$\begin{array}{ccc} & X & \\ & {\scriptstyle C}\swarrow \quad \searrow{\scriptstyle D} & \\ \mathrm{const}_c & \Longrightarrow & \mathrm{const}_d \end{array}$$

Therefore for any diagram $X\colon \mathsf{J} \to \mathsf{C}$ we have a category of cocones $X \downarrow \Delta$.

Notation. For a diagram $X\colon \mathsf{J} \to \mathsf{C}$ we will denote a cone $C\colon \mathrm{const}_c \Rightarrow X$ by

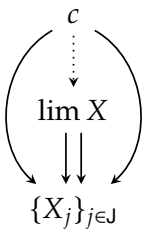
$$\begin{array}{c} c \\ \downarrow\downarrow \\ \{X_j\}_{j\in\mathsf{J}} \end{array}$$

and a cocone $D\colon X \Rightarrow \mathrm{const}_d$ by

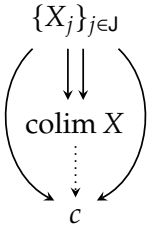
$$\begin{array}{c} \{X_j\}_{j\in\mathsf{J}} \\ \downarrow\downarrow \\ d \end{array}$$

3.2 Limits and colimits

Definition 3.2.1 (Limit)**.** A **limit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$ is a terminal object in the category of cones $\Delta \downarrow X$.
Less abstractly, a limit is a cone $\lim X\colon \mathrm{const}_{\lim X} \Rightarrow X$ such that for any other cone $C\colon \mathrm{const}_c \Rightarrow X$ there exists a unique morphism $c \to \lim X$ making the following diagram commute



Definition 3.2.2 (Colimit). A **colimit** of a diagram $X\colon \mathsf{J} \to \mathsf{C}$, denoted $\operatorname{colim} X$, is a terminal object in the category of cocones $X \downarrow \Delta$.
Less abstractly, a colimit is a cocone $\operatorname{colim} X\colon X \Rightarrow \operatorname{const}_{\operatorname{colim} X}$ such that for any other cocone $C\colon X \Rightarrow \operatorname{const}_c$ there exists a unique morphism $\operatorname{colim} X \to c$ making the following diagram commute



Proposition 3.2.1 (Uniqueness of limits and colimits). If a diagram $X\colon \mathsf{J} \to \mathsf{C}$ has a limit/colimit, it is unique up to unique isomorphism.

Proof. Direct from Proposition 1.3.1. □

Definition 3.2.3 (Functors and limits). Let $F\colon \mathsf{C} \to \mathsf{D}$ be a functor and $X\colon \mathsf{J} \to \mathsf{C}$ a diagram. We say that

- F **preserves** limits if whenever (L, φ) is the limit of X in C, $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D.
- F **reflects** limits if whenever (L, φ) is a cone over X such that $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D then (L, φ) is the limit of X in C.

Replacing cones with cocones and limits with colimits we get dual statements.

3.2.1 Examples of limits and colimits

We now define some special cases of limits and colimits that are generally useful. However, it is important to take into account that this constructions do not exist in general for all categories.

Products and coproducts

Let J be a small and discrete category.

Since the only morphisms in J are the identity morphisms, a diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists in a collection of objects $\{X_j\}_{j\in\mathsf{J}}$.

Therefore a cone with apex c over X is an object $c \in \mathsf{C}$ together with morphisms $c \to X_j$ for each $j \in \mathsf{J}$ and a cocone with nadir $c \in \mathsf{C}$ over X is an object c together with morphisms $X_j \to c$ for each $j \in \mathsf{J}$.

Note that any collection of objects $\{X_j\}_{j\in\mathsf{J}}$ in C may be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Definition 3.2.4 (Product). Let $\{X_j\}_{j\in \mathsf{J}}$ be a collection of objects in a category C.
We define the **product** of $\{X_j\}_{j\in \mathsf{J}}$, denoted $\prod_{j\in \mathsf{J}} X_j$, as the limit of the diagram $X\colon \mathsf{J}\to \mathsf{C}$.
Less abstractly, a **product** is an object $\prod_{j\in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\pi_k\colon \prod_{j\in \mathsf{J}} X_j \to X_k$ for every $j\in \mathsf{J}$ such that, for any map $Y\to X_j$ there exists a unique morphism $Y\to \prod_{j\in \mathsf{J}} X_j$ making the following diagram commute

$$\begin{array}{ccc} Y & \dashrightarrow & \prod_{j\in \mathsf{J}} X_j \longrightarrow X_j \\ & \searrow & \uparrow \\ & & \end{array}$$

Definition 3.2.5 (Coproduct). Let $\{X_j\}_{j\in \mathsf{J}}$ be a collection of objects in a category C.
We define the **coproduct** of $\{X_j\}_{j\in \mathsf{J}}$, denoted $\coprod_{j\in \mathsf{J}} X_j$, as the colimit of the diagram $X\colon \mathsf{J}\to \mathsf{C}$.
Less abstractly, a **coproduct** is an object $\coprod_{j\in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\iota_k\colon X_k \to \coprod_{j\in \mathsf{J}} X_j$ for every $j\in \mathsf{J}$ such that, for any map $X_j\to Y$ there exists a unique morphism $\coprod_{j\in \mathsf{J}} X_j \to Y$ making the following diagram commute

$$\begin{array}{ccc} Y & \dashleftarrow & \coprod_{j\in \mathsf{J}} X_j \longleftarrow X_j \\ & \nwarrow & \downarrow \\ & & \end{array}$$

Remark 3.2.1. Consider the case where $\mathsf{J}=\varnothing$ is the empty category, i.e: $\mathrm{Ob}(\mathsf{J})=\varnothing$.
Suppose that products exist in C.
Let's see how are $\varnothing$-shaped products (empty products).
Note that the family of objecs $\{X_j\}_{j\in\varnothing}\subseteq \mathsf{C}$ is empty, and so there are no projection maps $\pi_k : \prod_{j\in\varnothing} X_j \to X_k$ nor morphisms $\{f_j : Y\to X_j\}_{j\in\varnothing}$ for all $Y\in\mathsf{C}$. Therefore the product condition is vacuosly true, i.e: there exists a unique map $Y\to \prod_{j\in\varnothing} X_j$ for all $Y\in\mathsf{C}$. This means that $\prod_{j\in\varnothing} X_j$ is a terminal object in C.
Dually, if C has coproducts then the empty coproduct is initial in C.

Equalisers and coequalisers

Consider the *parallel pair* category

$$\mathsf{J} = \quad \bullet \rightrightarrows \bullet$$

A diagram $X\colon \mathsf{J}\to\mathsf{C}$ corresponds to a pair of morphisms $f,g\colon A\rightrightarrows B$ in C.

Therefore a cone $C\colon \mathrm{const}_c \Rightarrow X$ is a morphism $c\colon C\to A$ such that $fc=gc$:

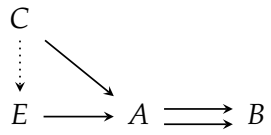
$$C \xrightarrow{c} A \underset{g}{\overset{f}{\rightrightarrows}} B$$

Analogously a cocone $D\colon X\Rightarrow \mathrm{const}_d$ is a morphism $d\colon B\to D$ such that $df=dg$:

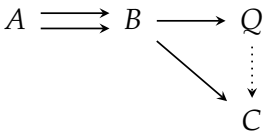
$$A \underset{g}{\overset{f}{\rightrightarrows}} B \xrightarrow{d} D$$

Note that any pair of morphisms $f,g\colon A\rightrightarrows B$ can be considered as a diagram $X\colon \mathsf{J}\to\mathsf{C}$.

Definition 3.2.6 (Equaliser). Let $f,g\colon A\rightrightarrows B$ be two morphisms in C, considered as a diagram $X\colon \mathsf{J}\to\mathsf{C}$. The **equaliser** of f and g is the limit of this diagram.
Less abstractly, it is an object $E\in\mathsf{C}$ together with a morphism $\mathrm{eq}\colon E\to A$ such that for any morphism $c\colon C\to A$ with $fc=gc$ there exists a unique map $C\to E$ the following diagram commute



Definition 3.2.7 (Coequaliser). Let $f, g : A \rightrightarrows B$ be two morphisms in C, considered as a diagram $X : \mathsf{J} \to \mathsf{C}$. The **coequaliser** of f and g is the limit of this diagram. Less abstractly, it is an object $Q \in \mathsf{C}$ together with a morphism $\mathrm{coeq} : B \to Q$ such that for any morphism $d : B \to C$ with $df = dg$ there exists a unique map $Q \to C$ making the following diagram commute



Proposition 3.2.2. Equaliser arrows are monomorphisms. Dually, coequaliser arrows are epimorphisms.

Proof. Let $\mathrm{eq} : E \to X$ be a equaliser. Consider two morphisms $f, g : C \rightrightarrows E$ in C such that $\mathrm{eq} \circ f = \mathrm{eq} \circ g$. By the uniqueness of the map $\mathrm{eq} : E \to X$ we have $f = g$. □

Example 3.2.1 (Equalisers and coequalisers in Set). In Set the equaliser of two maps $f, g : A \rightrightarrows B$ is $E = \{a \in A \mid f(a) = g(a)\}$ together with the inclussion $E \hookrightarrow A$. The coequaliser of f and g is $Q = {}^{B}\!/_{\sim}$ where $\sim$ is the smallest equivalence relation such that for all $a \in A \quad f(a) \sim g(a)$ together with the natural projection

$$\begin{aligned} \mathrm{coeq} : B &\twoheadrightarrow Q \\ b &\mapsto [b] \end{aligned}$$

A special case of equalisers and coequalisers are kernels and cokernels.

Definition 3.2.8 (Kernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **kernel** $\ker f$ of a morphism $f : A \to B$ is the equaliser

$$\ker f \rightarrowtail A \underset{0}{\overset{f}{\rightrightarrows}} B$$

Less abstractly, it is an object $\ker f$ together with a morphism $i : \ker f \to X$ such that for every $C \in \mathsf{C}$ and every morphism $g : C \to X$ such that $f \circ g = 0$ there is a unique morphism $\phi : C \to \ker f$ such that $g = p \circ \phi$

Definition 3.2.9 (Cokernel). Let C be a category with a zero object $0 \in \mathsf{C}$. The **cokernel** $\mathrm{coker}\, f$ of a morphism $f : A \to B$ is the coequaliser

$$A \underset{0}{\overset{f}{\rightrightarrows}} B \twoheadrightarrow \mathrm{coker}\, f$$

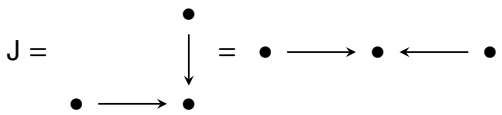
Less abstractly, it is an object $\mathrm{coker}\, f$ together with a morphism $p : Y \to \mathrm{coker}\, f$ such that for every $C \in \mathsf{C}$ and every morphism $g : Y \to C$ such that $g \circ f = 0$ there is a unique morphism $\phi : \mathrm{coker}\, f \to C$ such that $g = \phi \circ p$.
We denote

$${}^{A}\!/_{B} := \mathrm{coker}(A \to B)$$

Remark 3.2.2. By Proposition 3.2.2, kernels are monomorphisms and cokernels are epimorphisms.

Pullbacks and pushouts

Consider the category



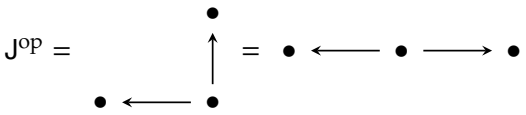
A diagram $X\colon \mathsf{J} \to \mathsf{C}$ consists of a pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$.

Therefore a cone $\mathrm{const}_C \Rightarrow X$ consists of an object $C \in \mathsf{C}$ together with morphisms $C \to X$ and $C \to Y$ making the following diagram commute.

$$\begin{array}{ccc} C & \longrightarrow & A \\ \downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow[g]{} & Z \end{array}$$

Note that any pair of morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.

Consider now the opposite category



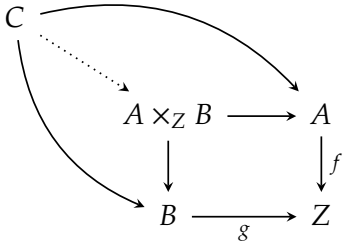
A diagram $Y\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$ consists of a pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$.

Therefore a cocone $X \Rightarrow \mathrm{const}_D$ consists of an object $D \in \mathsf{D}$ together with morphisms $A \to D$ and $B \to D$ making the following diagram commute

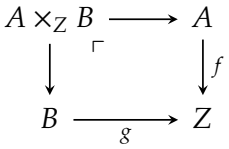
$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

Note that any pair of morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ can be considered as a diagram $X\colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.

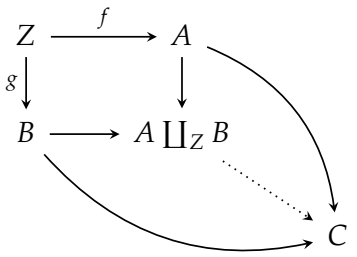
Definition 3.2.10 (Pullback). Let $A \xrightarrow{f} Z \xleftarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X\colon \mathsf{J} \to \mathsf{C}$.
The **pullback** (or **fibre product**) of f and g is the limit of this diagram, denoted by $A \times_Z B$.
Less abstractly, it is an object $A \times_Z B \in \mathsf{C}$ together with a pair of morphisms $\pi_A\colon A \times_Z B \to A, \pi_B : A \times_Z B \to B$ such that for any pair of morphisms $C \to A, C \to B$ making the square commute, there exists a unique morphism $C \to A \times_Z B$ such that



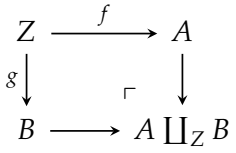
Pullback diagrams are often denoted by



Definition 3.2.11 (Pushout). Let $A \xleftarrow{f} Z \xrightarrow{g} B$ be a pair of morphisms in C, considered as a diagram $X \colon \mathsf{J}^{\mathrm{op}} \to \mathsf{C}$.
The **pushout** of f and g is the colimit of this diagram, denoted by $X \coprod_Z Y$.
Less abstractly, it is an object $A \coprod_Z B \in \mathsf{C}$ together with a pair of morphisms $\iota_A \colon A \to A \coprod_Z B, \iota_B : B \to A \coprod_Z B$ such that for any pair of morphisms $A \to C, B \to C$ making the square commute, there exists a unique morphism $A \coprod_Z B \to C$ such that



Pushout diagrams are often denoted by



Example 3.2.2 (Pullbacks and pushouts in Set). In Set the pullback of two morphisms $A \xrightarrow{f} Z \xleftarrow{g} B$ is the set

$$A \times_Z B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$$

together with the projection maps

$$\begin{aligned} \pi_A : A \times_Z B &\to A \\ (a, b) &\mapsto a \end{aligned} \qquad \begin{aligned} \pi_B : A \times_Z B &\to B \\ (a, b) &\mapsto b \end{aligned}$$

The pushout of two morphisms $A \xleftarrow{f} Z \xrightarrow{g} B$ is the set

$$A \coprod_Z B = {}^{A \coprod B}\!/_{\sim}$$

where $\sim$ is the equivalence relation generated by $f(a) \sim g(a)$ for all $a \in A$ together

the inclusion maps

$$\iota_A : A \to A \coprod_Z B \qquad \iota_B : B \to A \coprod_Z B$$
$$a \mapsto [a, 0] \qquad b \mapsto [b, 1]$$

Remark 3.2.3. Notice how pullbacks in Set can be seen as a equaliser in Set from the product maps $f \circ \pi_A, g \circ \pi_B : A \times B \to Z$. Considering pushouts in Set we get a dual observation.
Indeed, it is not difficult to check that if C has products then the pullback

$$\begin{array}{ccc} A \times_Z B & \longrightarrow & A \\ \downarrow & \ulcorner & \downarrow f \\ B & \xrightarrow[g]{} & Z \end{array}$$

is equivalent to the equaliser

$$A \times_Z B \longrightarrow A \times B \underset{g\circ\pi_B}{\overset{f\circ\pi_A}{\rightrightarrows}} Z$$

Dually if C has coproducts the pushout

$$\begin{array}{ccc} Z & \xrightarrow{f} & A \\ {\scriptstyle g}\downarrow & \ulcorner & \downarrow \\ B & \longrightarrow & A \coprod_Z B \end{array}$$

is equivalent to the coequaliser

$$Z \underset{\iota_B\circ g}{\overset{\iota_A\circ f}{\rightrightarrows}} A \coprod B \longrightarrow A \coprod_Z B$$

For a proof see [Rie, Lemmas 3.4.16, 3.4.17].

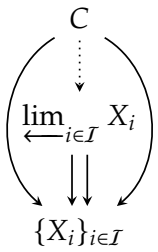
Inverse and direct limits

Definition 3.2.12 (Directed set)**.** A **directed set** is a preordered set $(\mathcal{I}, \leq)$ such that for every $i, j \in \mathcal{I}$, there exists $k \in \mathcal{I}$ with $i \leq k$ and $j \leq k$. We can view this as a category following Example 1.1.5.

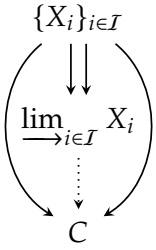
Example 3.2.3. Let $(X, \mathcal{T})$ be a topological space. Then $(\mathcal{T}, \subseteq)$ forms a directed set. This means that the category of open sets $\mathsf{Open}(X)$ can be seen as a directed set.

Definition 3.2.13 (Direct and inverse systems)**.** A **direct system** in C over $\mathcal{I}$ is a diagram $X : \mathcal{I} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i\in\mathcal{I}}$ and morphisms $f_{ij} : X_i \to X_j$ for all $i \leq j$, such that $f_{ii} = \mathrm{id}_{X_i}$ and $f_{jk} \circ f_{ij} = f_{ik}$ for all $i \leq j \leq k$.
Dually, an **inverse system** in C over $\mathcal{I}$ is a diagram $X : \mathcal{I}^{\mathrm{op}} \to \mathsf{C}$. Less abstractly, it consists of a family of objects $\{X_i\}_{i\in\mathcal{I}}$ and morphisms $p_{ij} : X_j \to X_i$ for all $i \leq j$, such that $p_{ii} = \mathrm{id}_{X_i}$ and $p_{ij} \circ p_{jk} = p_{ik}$ for all $i \leq j \leq k$.

Definition 3.2.14 (Inverse limit)**.** The **inverse limit** (or **projective limit**) of an inverse system $X : \mathcal{I} \to \mathsf{C}$ is the limit of this diagram, denoted as $\varprojlim_{i\in\mathcal{I}} X_i$.
Less abstractly, it is an object $\varprojlim_{i\in\mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\pi_i : \varprojlim_{i\in\mathcal{I}} X_i \to X_i$ such that $p_{ij} \circ \pi_j = \pi_i$ for all $i \leq j$. For any other object C and morphisms $c_i : C \to X_i$ compatible with the system ($p_{ij} \circ c_j = c_i$), there exists a unique morphism $C \to \varprojlim_{i\in\mathcal{I}} X_i$ making the following diagram commute:



Definition 3.2.15 (Direct limit)**.** The **direct limit** (or **inductive limit**) of a direct system $X\colon \mathcal{I} \to \mathsf{C}$ is the colimit of this diagram, denoted as $\varinjlim_{i\in\mathcal{I}} X_i$.
Less abstractly, it is an object $\varinjlim_{i\in\mathcal{I}} X_i \in \mathsf{C}$ together with morphisms $\iota_i\colon X_i \to \varinjlim_{i\in\mathcal{I}} X_i$ such that $\iota_j \circ f_{ij} = \iota_i$ for all $i \leq j$. For any other object C and morphisms $c_i\colon X_i \to C$ compatible with the system ($c_j \circ f_{ij} = c_i$), there exists a unique morphism $\varinjlim_{i\in\mathcal{I}} X_i \to C$ making the following diagram commute:



Example 3.2.4 (Inverse limits in Set)**.** In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{p_{ij}\}_{i\leq j})$ is an inverse system indexed by a directed set $(\mathcal{I}, \leq)$, its inverse limit is

$$\varprojlim_{i\in\mathcal{I}} X_i = \left\{ (x_i)_{i\in\mathcal{I}} \in \prod_{i\in\mathcal{I}} X_i \mid \forall i \leq j \quad x_i = p_{ij}(x_j) \right\}$$

together with the natural projection maps $\pi_k((x_i)_{i\in\mathcal{I}}) = x_k$.

Example 3.2.5 (Direct limits in Set)**.** In Set, if $(\{X_i\}_{i\in\mathcal{I}}, \{f_{ij}\}_{i\leq j})$ is a direct system indexed by a directed set $(\mathcal{I}, \leq)$, its direct limit is

$$\varinjlim_{i\in\mathcal{I}} X_i = \coprod_{i\in\mathcal{I}} X_i \Big/ \sim$$

where $\sim$ is the equivalence relation defined by $x_i \sim x_j$ if and only if there exists $k \in \mathcal{I}$ with $k \geq i, j$ such that

$$f_{ik}(x_i) = f_{jk}(x_j)$$

together with the natural insertion maps $\iota_k(x_k) = [x_k]$.

We finish this chapter by giving a condition for the existence of finite limits and colimits

Theorem 3.2.1. A category has all finite limits if and only if it has a terminal object, binary products and equalisers.
Dually it has all finite colimits if and only if it has a initial object, binary coproducts and coequalisers.

Proof.

$\Longrightarrow$ Direct since all of these are particular cases of limits.

$\Longleftarrow$ Consider a diagram $X\colon \mathsf{J} \Rightarrow \mathsf{C}$.
Since binary products exist in C we can construct inductively

$$P \coloneqq \prod_{j\in\mathsf{J}} X_j \in \mathsf{C}$$

For every morphism $u\colon j \to k$ in J we can consider two maps from P to X_k: the projection $\pi_k\colon P \to X_k$ and the composition $Xu \circ \pi_j\colon P \to X_j \to X_k$. Therefore we can consider a pair of parallel morphisms

$$\prod_{j\in\mathsf{J}} X_j \underset{t}{\overset{s}{\rightrightarrows}} \prod_{u:j\to k\in\mathsf{J}} X_k$$

where s is defined in each component as π_k and t is $Xu \circ \pi_j$ in each component. The equaliser of s and t (which exists by hypothesis) is then the limit of X. Indeed, let $\mathrm{eq}\colon L \to \prod_{j\in\mathsf{J}} X_j$ be this equaliser. This comes with a set of morphisms $\{\phi_j\colon L \to X_j\}_{j\in\mathsf{J}}$ such that

$$\phi_j = \pi_j \circ \mathrm{eq}$$

By definition $s \circ \mathrm{eq} = t \circ \mathrm{eq}$, so for every $u\colon j \to k$ in J we have

$$\pi_k \circ \mathrm{eq} = (Xu \circ \pi_j) \circ \mathrm{eq}$$

Therefore

$$\phi_k = Xu \circ \phi_j$$

and so L is a cone over X.
To see that it is a limit cone consider another cone $(W, \{\psi_j\colon W \to X_j\})$ such that for any $u\colon j \to k$ we have $\psi_k = Xu \circ \psi_j$. By definition of the product there exists a unique morphism $h\colon W \to \prod_{j\in\mathsf{J}} W_j$ such that $\pi_j \circ h = \psi_j$.
Note that $s \circ h = t \circ h$. Indeed the components of $s \circ h$ are $\pi_k \circ h = \psi_k$ and the components of $t \circ h$ are $Xu \circ \pi_j \circ h = Xu \circ \psi_j = \psi_k$.
Therefore by the definition of equaliser there exists a unique morphism $\widetilde{h}\colon W \to L$ such that $\mathrm{eq} \circ \widetilde{h} = h$. Finally since for any $j \in \mathsf{J}$

$$\phi_j \circ \widetilde{h} = \pi_j \circ \mathrm{eq} \circ \widetilde{h} = \pi_j \circ h = \psi_j$$

we have that W factors uniquely through L and so L is the limit of X.

□

Chapter 4

More on functors

4.1 Representable functors

Definition 4.1.1. A functor $F : \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists an object $X \in \mathsf{C}$ such that

$$F \cong \mathrm{Hom}_{\mathsf{C}}(X, -)$$

where $\mathrm{Hom}_{\mathsf{C}}(X, -)$ is the covariant Hom functor.

Example 4.1.1. The covariant Hom functor is representable.

Definition 4.1.2 (Concrete category)**.** A category C is **concrete** if there exists a faithful functor $U : \mathsf{C} \to \mathsf{Set}$. Such a functor is often called a **forgetful functor**.

Theorem 4.1.1 (Yoneda - [Rie, Theorem 2.2.4])**.** Let C be a category, $X \in \mathsf{C}$ an object, and $F : \mathsf{C} \to \mathsf{Set}$ a functor. There is a bijection:

$$\Phi : \mathrm{Hom}(\mathrm{Hom}(X, -), F) \xrightarrow{\sim} F(X)$$

given by $\alpha \mapsto \alpha_X(\mathbb{1}_X)$. This bijection is natural in both X and F.

4.2 Adjunctions

Definition 4.2.1 (Adjoint functors)**.** Let $L : \mathsf{C} \to \mathsf{D}$ and $R : \mathsf{D} \to \mathsf{C}$ be a pair of functors. We say that (L, R) is a pair of **adjoint functors** or that L is left-adjoint to R and R is right-adjoint to L if there exists a natural isomorphism of bifunctors

$$\mathrm{Hom}_{\mathsf{D}}(L(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(-, R(-)) \tag{4.2.1}$$

We represent this diagramatically using

$$\mathsf{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathsf{D} \quad (\perp)$$

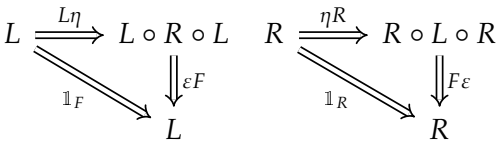
Remark 4.2.1 (Unit and counit of an adjunction)**.** Let (L, R) be a pair of adjoint functors. The natural isomorphism (4.2.1) gives the natural isomorphisms

$$\mathrm{Hom}_{\mathsf{D}}(L(-), L(-)) \cong \mathrm{Hom}_{\mathsf{C}}(-, (G \circ L)(-))$$
$$\mathrm{Hom}_{\mathsf{D}}((L \circ R)(-), -) \cong \mathrm{Hom}_{\mathsf{C}}(R(-), R(-))$$

In particular we get natural transformations

$$\eta : \mathbb{1}_{\mathsf{C}} \Rightarrow R \circ L \qquad \varepsilon : L \circ R \Rightarrow \mathbb{1}_{\mathsf{D}}$$

that satisfy the *triangular identities*



Conversely, if L, R are two pair of functors and ε, η are natural transformations as above then (L, R) form an adjoint pair. For a proof of this see [**riehlCategoryTheory**].

Remark 4.2.2. Left and right adjoints are dual concepts.

Proposition 4.2.1 (Adjoints are unique)**.** The right adjoint (resp. left adjoint) to a functor, provided it exists, is unique up to natural isomorphism

Proof. Let $L : \mathsf{C} \to \mathsf{D}$ be a functor.

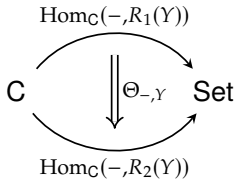
Let $R_1, R_2 : \mathsf{D} \to \mathsf{C}$ be two functors which are right adjoint to L. This means that for all $X \in \mathsf{C}$ and $Y \in \mathsf{D}$ we have bijections

$$\phi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_1(Y))$$
$$\psi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Using this we can define a bijection

$$\Theta_{X,Y} = \psi_{X,Y} \circ \phi_{X,Y}^{-1} : \mathrm{Hom}_{\mathsf{C}}(X, R_1(Y)) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Since ψ and ϕ are natural in X so is Θ. Therefore for a fixed $Y \in \mathsf{D}$ we have a natural isomorphism



By the Yoneda lemma $\Theta_{-,Y}$ induces a unique isomorphism in C

$$\alpha_Y : R_1(Y) \xrightarrow{\sim} R_2(Y)$$

which ensemble into a natural isomorphism

$$\alpha : R_1 \Rightarrow R_2$$

□

Theorem 4.2.1 ([Rie, Theorem 4.5.2])**.** Right adjoints preserve all limits that exist in their codomain.
Dually left adjoints preserve all colimits that exist in their domain.

4.3 Functor categories

Definition 4.3.1 (Functor category). Let C and D be categories. The **functor category** $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has:

- **Objects**: functors $F : \mathsf{C} \to \mathsf{D}$.
- **Morphisms**: natural transformations $\eta : F \implies G$.

Composition and identities are defined componentwise.

Proposition 4.3.1. Let C and D be two categories and $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ be its functor category. If D has limits (resp. colimits) of shape J then $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$ has limits (resp. colimits) of shape J. Furthermore, they are computed objectwise.

Proof. Consider a J-shaped diagram $X : \mathsf{J} \Rightarrow [\mathsf{C}, \mathsf{D}]$. We wish to construct a limit object $L \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ together with a limit cone $\pi_j : L \to X_j$ for all $j \in \mathsf{J}$.

For any object $c \in \mathsf{C}$ the *evaluation* functor

$$\begin{array}{rccc} \mathrm{ev}_c : & [\mathsf{C}, \mathsf{D}] & \longrightarrow & \mathsf{D} \\ & F & & F(c) \\ & \eta \Downarrow & \longmapsto & \downarrow \eta_c \\ & G & & G(c) \end{array}$$

allows us to define a diagram $X^c := \mathrm{ev}_c \circ X : J \Rightarrow \mathsf{D}$. By hypothesis there exists

$$\lim_{j \in \mathsf{J}} X^c(j) =: L(c) \qquad \pi^c_j : L(c) \to X^c(j)$$

To make L into a functor we need to define its action on morphisms. Consider a morphism $f : c \to d$ in C. Since X_j is a functor for all $j \in \mathsf{J}$ we have an induced morphism $X_j f : X_j(c) \to X_j(d)$. Let's consider the family of morphism

$$\left\{ X_j f \circ \pi^c_j : L(c) \to X_j(d) \right\}_{j \in \mathsf{J}}$$

We want to show that this forms a cone over $L(d)$ with apex $L(c)$

Let $u : j \to k$ be a morphism in J. Since $Xu : X_j \Rightarrow X_k$ is a natural transformation we have a commutative diagram

$$\begin{array}{ccc} X_j(c) & \xrightarrow{(Xu)_c} & X_k(x) \\ {\scriptstyle X_j f}\downarrow & & \downarrow{\scriptstyle X_k f} \\ X_j(d) & \xrightarrow[(Xu)_d]{} & X_k(d) \end{array}$$

Since π^c is a limit over X^c we also now that $(Xu)_c \circ \pi^c_j = \pi^c_k$. Therefore we have

$$(Xu)_d \circ (X_j f \circ \pi^c_j) = ((Xu)_d \circ X_j f) \circ \pi^c_j = X_k f \circ (Xu)_c \circ \pi^c_j = X_k f \circ \pi^c_k$$

i.e: it is a valid cone over $L(d)$. By the universal property of $L(d) = \lim_{j \in \mathsf{J}} X^d(j)$ there exists a unique morphism $Lf : L(c) \to L(d)$ such that for all $j \in \mathsf{J}$ the following diagram commutes

$$\begin{array}{ccc} L(c) & \xrightarrow{Lf} & L(d) \\ {\scriptstyle \pi_j^c}\downarrow & & \downarrow{\scriptstyle \pi_j^d} \\ X^c(j) & \xrightarrow[X_j f]{} & X^d(j) \end{array}$$

The functiorality of L follows directly from the uniqueness of this induced map into $L(d)$. Moreover the above commutative square is the naturallity condition for π_j with respect to f. Therefore, for each $j \in \mathsf{J}$ the collection $\{\pi_j^c\}_{c \in \mathsf{C}}$ is a natural transformation $\pi_j : L \Rightarrow X_j$. It is an easy check to see that this collection forms a cone over X in the functor category $\mathsf{Fct}(\mathsf{C}, \mathsf{D})$.

To finish the proof, we need to check the universal property. Consider another functor $M \in \mathsf{Fct}(\mathsf{C}, \mathsf{D})$ equipped with a cone $\tau_j : M \Rightarrow X_j$.

For each $c \in \mathsf{C}$ the evaluation $\tau_j^c := \mathrm{ev}_c \circ M : M(c) \to X^c(j)$ is a cone over X^c. This induces a unique morphism $\phi_c : M(c) \to L(c)$ such that $\pi_j^c \circ \phi_c = \tau_j^c$.

To show that $\phi \equiv \{\phi_c\}_{c \in \mathsf{C}}$ is a natural transformation let $f : c \to d$ be a morphism in C. We want to see that the following diagram commutes

$$\begin{array}{ccc} M(c) & \xrightarrow{\phi_c} & L(c) \\ {\scriptstyle Mf}\downarrow & & \downarrow{\scriptstyle Lf} \\ M(d) & \xrightarrow[\phi_d]{} & L(d) \end{array}$$

Postcompossing with π_j^d we get, on the one hand

$$\pi_j^d \circ (Lf \circ \phi_c) = (\pi_j^d \circ Lf) \circ \phi_c = X_j f \circ \pi_j^c \circ \phi_c = X_j f \circ \tau_j^c$$

And on the other

$$\pi_j^d \circ (\phi_d \circ Mf) = (\pi_j^d \circ \phi_d) \circ Mf = \tau_j^d \circ Mf$$

Since τ_j is a natural transformation we have

$$X_j f \circ \tau_j^c = \tau_j^d \circ Mf$$

Therefore since the projections agree for all $j \in \mathsf{J}$ by uniqueness of the maps π_j we get the commutativity of the diagram. Its uniqueness as a morphism of cones follows directly from the objectwise uniqueness of each ϕ_c. □

PART II

Homological algebra

Chapter 5

Additive categories

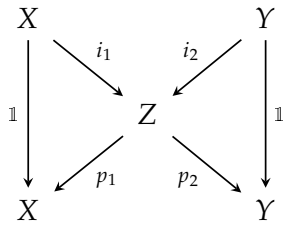
Definition 5.0.1 (Additive category). A category C is **additive** if it satisfies

- **Ad.1)** For every pair of objects $X, Y \in \mathsf{C}$ the set $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.
- **Ad.2)** There exists a zero object $0 \in \mathsf{C}$.
- **Ad.3)** C admits finite coproducts.
- **Ad.4)** C admits finite products.

Notation. If $X, Y \in \mathsf{C}$ we denote by $X \oplus Y$ its coproduct and call it their **direct sum**.

Lemma 5.0.1. Let C be a category satisfying Ad.1 and Ad.2.
Suppose that C verifies

- **Ad.5)** For any pair of objects $X, Y \in \mathsf{C}$ there exists $Z \in \mathsf{C}$ and morphisms



such that

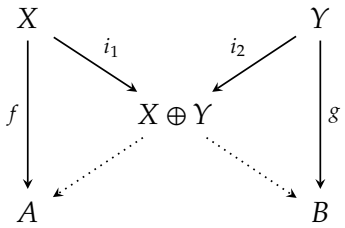
$$p_1 \circ i_1 = \mathbb{1}_X \quad p_1 \circ i_2 = 0$$
$$p_2 \circ i_2 = \mathbb{1}_Y \quad p_2 \circ i_1 = 0$$
$$i_1 \circ p_1 + i_2 \circ p_2 = \mathbb{1}_Z$$

Then conditions Ad.4, Ad.3 and Ad.5 are equivalent and the objects $X \times Y$, $X \oplus Y$ and Z are naturally isomorphic.

Proof. We will show that conditions Ad.3 and Ad.5 are equivalent and that $X \oplus Y \cong Z$. A dual argument shows that Ad.4 and Ad.5 are equivalent and that $X \times Y \cong Z$.

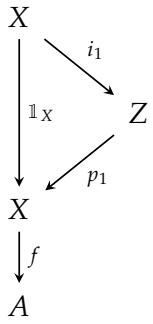
Ad.3 $\implies$ Ad.5 Let $X \oplus Y$ be the coproduct of X and Y, that comes with morphisms $i_1 \colon X \to X \oplus Y$ and $i_2 \colon Y \to X \oplus Y$.

By definition, for all morphisms $f\colon X \to A$, $g\colon Y \to B$ there exist unique morphisms $X \oplus Y \to A$, $X \oplus Y \to B$ such that



In particular if $f = \mathbb{1}_X$ and $g = \mathbb{1}_Y$

Ad.5 $\implies$ Ad.3 We are going to see that Z verifies the universal property of the coproduct. Let $f : X \to A$ be a morphism. We want to find a morphism $\widetilde{f}\colon Z \to A$ such that $f = \widetilde{f} \circ i_1$. We have the following



Therefore if $\widetilde{f} := f \circ p_1$ since $i_1 \circ p_1 = \mathbb{1}_X$ we have $\widetilde{f} \circ i_1 = f$.
Analogously we see that any morphism $g : Y \to B$ can can be factored through Z. Therefore since Z verifies the universal property of the coproduct, $Z \cong X \oplus Y$.

□

Lemma 5.0.2. A morphism $f\colon X \to Y$ in an additive category is a monomorphism if and only if for every morphism $g\colon Z \to X$

$$f \circ g = 0 \implies g = 0$$

Dually f is an epimorphism if and only if for every morphism $g\colon Y \to Z$

$$g \circ f = 0 \implies g = 0$$

Proof. We will check the monomorphism caracterization.

$\implies$ Let $f\colon X \rightarrowtail Y$ be a monomorphism. Consider a morphism $g\colon Z \to X$ such that $f \circ g = 0$. Since $f \circ 0 = 0$ applying left-cancellativity we have that $g = 0$.

$\impliedby$ Let $g_1, g_2\colon Z \rightrightarrows X$ be two morphisms such that $f \circ g_1 = f \circ g_2$.
Since $\mathrm{Hom}(Z, Y)$ is an abelian group we have

$$0 = f \circ g_1 - f \circ g_2 = f \circ (g_1 - g_2)$$

Therefore by hypothesis $g_1 - g_2 = 0 \iff g_1 = g_2$, proving that f is a monomorphism.

□

Lemma 5.0.3. Let I be a small category. If C is additive then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Proof.

Ad.1 For any two natural transformations $\phi, \psi \colon \mathsf{I} \Rightarrow \mathsf{C}$ we can define their addition componentwise: $(\phi + \psi)_i = \phi_i + \psi_i$ for all $i \in \mathsf{I}$. Since $\phi_i, \psi_i \in \mathrm{Hom}_{\mathsf{C}}(-,-)$ this is well defined.

Ad.2 Consider the zero functor

$$\begin{array}{ccc} 0 \colon \mathsf{I} & \longrightarrow & \mathsf{C} \\ X & & 0 \\ f \downarrow & \longmapsto & \downarrow 0 \\ Y & & 0 \end{array}$$

This is clearly the zero object in $\mathsf{Fct}(\mathsf{I}, \mathsf{C},)$.

Ad.3, Ad.4 Direct from Proposition 4.3.1.

□

Definition 5.0.2 (Additive functor). Let C, D be additive categories. A functor $F \colon \mathsf{C} \to \mathsf{D}$ is **additive** if for all $X, Y \in \mathsf{C}$

$$\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$$

is a morphism of abelian groups.

Proposition 5.0.1. Let C, D be additive categories. A functor $F \colon \mathsf{C} \to \mathsf{D}$ is additive if and only if it commutes with finite direct sums, i.e: if $F(0) \cong 0$ and for all $X, Y \in \mathsf{C}$

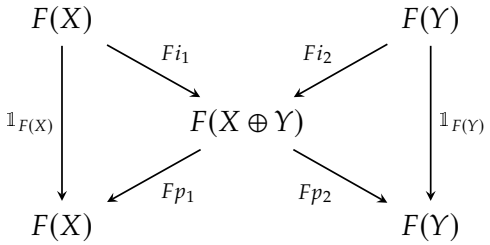
$$F(X \oplus Y) = F(X) \oplus F(Y)$$

Proof.

$\Longrightarrow$ Since F is a functor $F(\mathbb{1}_0) = \mathbb{1}_{F(0)}$. Furthermore since F is additive if $0_{X,Y}$ is the zero morphism between $X, Y \in \mathsf{C}$ then $F(0_{X,Y}) = 0_{F(X),F(Y)}$ because group homomorphisms preserve the identity. In particular taking $X = Y = 0$ since $0_{0,0} = \mathbb{1}_0$ we have

$$\mathbb{1}_{F(0)} = F(\mathbb{1}_0) = F(0_{0,0}) = 0_{F(0),F(0)}$$

which means that $F(0)$ is the zero object in D.
Let $X, Y \in \mathsf{C}$ be two objects and $X \oplus Y$ be its direct sum. Applying F to the diagram in Lemma 5.0.1 defining $Z \cong X \oplus Y$ we get



Note that

$$Fp_1 \circ Fi_1 = F(p_1 \circ i_1) = F(\mathbb{1}_X) = \mathbb{1}_{F(X)} \qquad Fp_1 \circ Fi_2 = F(p_1 \circ i_2) = F(0) = 0$$
$$Fp_2 \circ Fi_2 = F(p_2 \circ i_2) = F(\mathbb{1}_Y) = \mathbb{1}_{F(Y)} \qquad Fp_2 \circ Fi_1 = F(p_2 \circ i_1) = F(0) = 0$$

Furthermore since F is additive

$$Fp \circ Fi + Fj \circ Fq = F(p \circ i + j \circ q) = F\mathbb{1}_{X \oplus Y} = \mathbb{1}_{F(X \oplus Y)}$$

Therefore by Lemma 5.0.1 $F(X \oplus Y) \cong F(X) \oplus F(Y)$.

$\Longleftarrow$ Applying Lemma 5.0.1 to $X \oplus X$ and $Y \oplus Y$ we get maps $\Delta_X : X \to X \oplus X$ and $\nabla_Y : Y \oplus Y \to Y$ such that

$$p_1 \circ \Delta_X = \mathbb{1}_X = p_2 \circ \Delta_X$$
$$\nabla_Y \circ i_1 = \mathbb{1}_Y = \nabla_Y \circ i_2$$

Then for any pair of morphisms $f, g : X \to Y$ in D we have

$$f + g = \nabla_Y \circ (f \oplus g) \circ \Delta_X$$

where $f \oplus g$ is the unique morphism such that

$$p_k \circ (f \oplus g) \circ i_j = \begin{cases} f & j = k = 1 \\ g & j = k = 2 \\ 0 & \text{otherwise} \end{cases}$$

Applying F to this expression we get

$$F(f + g) = F(\nabla_Y \circ (f \oplus g) \circ \Delta_X) = F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X$$

Since by hypothesis F preserves finit direct sums

$$F\nabla_Y \circ F(f \oplus g) \circ F\Delta_X = \nabla_{F(Y)} \circ (Ff \oplus Fg) \oplus \Delta_{F(X)} = Ff + Fg$$

Therefore $F(f + g) = Ff + Fg$, which means that F is additive.

□

Proposition 5.0.2. Let $L \dashv R : \mathsf{C} \to \mathsf{D}$ be a pair of adjoint functors between additive categories. Then L is additive if and only if R is additive.

Proof. Since left adjoints preserve colimits and right adjoints preserve limits we have

$$L(X \oplus Y) = L(X) \oplus L(Y)$$
$$R(X \oplus Y) = R(X \times Y) = R(X) \times R(Y) = R(X) \oplus R(Y)$$

Note that if R is additive then $L(0) = 0$ since

$$\mathrm{Hom}_{\mathsf{C}}(L(0), 0) = \mathrm{Hom}_{\mathsf{D}}(0, R(0)) = \mathrm{Hom}_{\mathsf{D}}(0, 0)$$

and viceversa.

Therefore by Proposition 5.0.1 we have the result. □

Definition 5.0.3 (Full additive subcategory). Let C be an additive category.
A full subcategory D of C is a **full additive subcategory** if the inclussion functor $\iota : D \hookrightarrow \mathsf{C}$ is additive.

Remark 5.0.1. By Proposition 5.0.1 a full subcategory D of C is a full additive subcategory if and only if $0 \in \mathsf{D}$ and D is closed under finite direct sums.

5.1 Complexes

From now until the end of the chapter, let C be an additive category.

Definition 5.1.1 (Differential object). A **differential object** $(X^\bullet, \mathrm{d}_X^\bullet)$ in C is a sequence of objects $X^k \in \mathsf{C}$ and morphisms d_X^k for every $k \in \mathbb{Z}$:

$$\cdots \longrightarrow X^{k-1} \xrightarrow{\mathrm{d}_X^{k-1}} X^k \xrightarrow{\mathrm{d}_X^k} X^{k+1} \longrightarrow \cdots$$

A morphism of differential objects $f^\bullet \colon (X^\bullet, \mathrm{d}_X^\bullet) \to (Y^\bullet, \mathrm{d}_Y^\bullet)$ is a collection of morphisms $\{f^k \colon X^k \to Y^k\}_{k\in\mathbb{Z}}$ such that, for all $k \in \mathbb{Z}$, the following diagram commutes

$$\begin{array}{ccccccc} \cdots & \longrightarrow & X^k & \xrightarrow{\mathrm{d}_X^k} & X^{k+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^k} & & \downarrow{\scriptstyle f^{k+1}} & & \\ \cdots & \longrightarrow & Y^k & \xrightarrow[\mathrm{d}_Y^k]{} & Y^{k+1} & \longrightarrow & \cdots \end{array}$$

Therefore we get a category $\mathsf{Diff}(\mathsf{C})$ of differential objects in C. By abuse of notation we will often denote differential objects $(X^\bullet, \mathrm{d}_X^\bullet)$ as $X^\bullet$.

Remark 5.1.1. We can view the category of differential objects $\mathsf{Diff}(\mathsf{C})$ as the functor category $\mathsf{Fct}(\mathbb{Z}, \mathsf{C})$. Therefore by Lemma 5.0.3 $\mathsf{Diff}(\mathsf{C})$ is an additive category.

Remark 5.1.2. For all $n \in \mathbb{Z}$ there exists a fully faithful functor

$$\iota_n \colon \mathsf{C} \longrightarrow \mathsf{Diff}(\mathsf{C})$$

which sends an object $X \in \mathsf{C}$ to the differential object

$$\cdots \longrightarrow 0 \longrightarrow X \longrightarrow 0 \longrightarrow \cdots$$

and a morphism $f \colon X \to Y$ in C to

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle f} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & Y & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

Therefore we can identify an object $X \in \mathsf{C}$ as a differential object using ι_0. When the context is clear, we will directly write $X \in \mathsf{Diff}(\mathsf{C})$ instead of $\iota_0(X) \in \mathsf{Diff}(\mathsf{C})$.

Definition 5.1.2 (Complex). A **complex** in C is a differential object $(X^\bullet, \mathrm{d}_X^\bullet)$ such that for all $k \in \mathbb{Z}$

$$\mathrm{d}_X^{k+1} \circ \mathrm{d}_X^k = 0$$

We denote by $\mathsf{Comp}(\mathsf{C})$ the full additive subcategory of $\mathsf{Diff}(\mathsf{C})$ consisting of complexes in C.

Definition 5.1.3. A complex $X^\bullet$ is

1. **Bounded above** if $X^n = 0$ of $n \gg 0$.
2. **Bounded below** if $X^n = 0$ of $n \ll 0$.
3. **Bounded** if it is bounded above and below.

We denote by $\mathsf{Comp}^*(\mathsf{C})$ ($* = +, -, \mathrm{b}$) the full additive subcategory of $\mathsf{Comp}(\mathsf{C})$ consisting of bounded above (resp. bounded below, bounded) complexes. To emphasize that we are working with unbounded complexes we may write $\mathsf{Comp}^{\mathrm{ub}}(\mathsf{C}) := \mathsf{Comp}(\mathsf{C})$.

Remark 5.1.3. Note that for any $n \in \mathbb{N}$, the inclussion functor ι_n can be viewed as a functor

$$\iota_n : \mathsf{C} \longrightarrow \mathsf{Comp}^*(\mathsf{C})$$

Therefore we can identify $X \in \mathsf{C}$ with $\iota_0(X) \in \mathsf{Comp}^*(\mathsf{C})$.

5.2 Homotopy of complexes

Definition 5.2.1 (Homotopy).

1. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is **homotopic to zero** if for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ in C such that
$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$
Visually
$$\begin{array}{ccccccccc} \cdots & \longrightarrow & X^{p-1} & \longrightarrow & X^p & \longrightarrow & X^{p+1} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f^{p-1}} & \swarrow{\scriptstyle s^p} & \downarrow{\scriptstyle f^p} & \swarrow{\scriptstyle s^{p+1}} & \downarrow{\scriptstyle f^{p+1}} & & \\ \cdots & \longrightarrow & Y^{p-1} & \longrightarrow & Y^p & \longrightarrow & Y^{p+1} & \longrightarrow & \cdots \end{array}$$
2. Two morphisms $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are **homotopic** if $f^\bullet - g^\bullet$ is homotopic to zero. We denote this by $f^\bullet \sim g^\bullet$.
3. A complex $X^\bullet \in \mathsf{Comp}(\mathsf{C})$ is **homotopic to zero** if $\mathbb{1}_{X^\bullet} \sim 0$.
4. A morphism of complexes $f^\bullet : X^\bullet \to Y^\bullet$ is a **homotopy equivalence** if there exists $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **homotopy equivalent** and write $X^\bullet \sim Y^\bullet$.

Lemma 5.2.1. Let $F : \mathsf{C} \to \mathsf{D}$ be an additive functor. If $f^\bullet : X^\bullet \to Y^\bullet$ is homotopic to zero then $F f^\bullet$ is also homotopic to zero.

Proof. Since $f^\bullet$ is homotopic to zero for all $p \in \mathbb{Z}$ there exists a morphism $s^p : X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

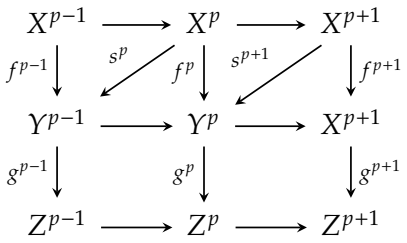
Therefore

$$F f^p = F(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p) = F s^{p+1} \circ \mathrm{d}_{F(X)}^p + \mathrm{d}_{F(Y)}^{p-1} \circ F s^p$$

□

Lemma 5.2.2. Let $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to Z^\bullet$ be two morphisms of complexes in $\mathsf{Comp}(\mathsf{C})$. If either f or g is homotopic to 0 then $g \circ f$ is homotopic to 0.

Proof. Assume that f is homotopic to 0. The proof consists on chasing the following diagram:



We have that for all $p \in \mathbb{Z}$ there exists $s^p : X^p \to Y^{p-1}$ such that

$$f^p = s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p$$

Therefore

$$\begin{aligned} g^p \circ f^p &= g^p \circ \left(s^{p+1} \circ \mathrm{d}_X^p + \mathrm{d}_Y^{p-1} \circ s^p \right) \\ &= \left(g^p \circ s^{p+1} \right) \circ \mathrm{d}_X^p + \left(g^p \circ \mathrm{d}_Y^{p-1} \right) \circ s^p \\ &= \left(g^p \circ s^{p+1} \right) \circ \mathrm{d}_X^p + \mathrm{d}_Z^{p-1} \circ \left(g^{p-1} \circ s^p \right) \end{aligned}$$

□

Chapter 6

Abelian categories

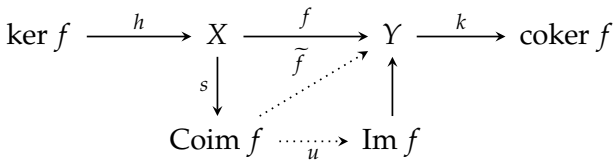
Definition 6.0.1. Let $f : X \to Y$ be a morphism in an additive category C that admits kernels and cokernels. Consider the following

$$\ker f \xrightarrow{h} X \xrightarrow{f} Y \xrightarrow{k} \operatorname{coker} f$$

We denote

$$\operatorname{Im} f := \ker k \qquad \operatorname{Coim} f := \operatorname{coker} h$$

I.e: we have the following diagram



Note that, since $f \circ h = 0$, f factors uniquely through $\operatorname{Coim} f$ producing the morphism $\widetilde{f}$. Furthermore since $k \circ f = k \circ \widetilde{f} = 0$ we have that $\widetilde{f}$ factors uniquely through $\operatorname{Im} f$. Therefore we have a canonical morphism

$$u : \operatorname{Coim} f \to \operatorname{Im} f$$

Definition 6.0.2 (Abelian category)**.** An additive category C is **abelian** if it verifies

Ab.1) Any morphism in C admits a kernel and a cokernel.

Ab.2) For any morphism f in C the canonical morphism $\operatorname{Coim} f \to \operatorname{Im} f$ is an isomorphism.

Lemma 6.0.1. An abelian category admits all finite limits and colimits.

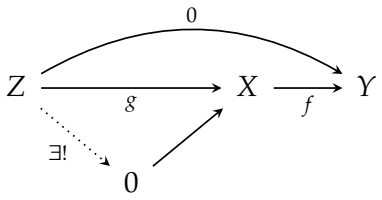
Proof. Direct from Theorem 3.2.1. □

Lemma 6.0.2. Let $f : X \to Y$ be a morphism in an abelian category. Then f is a monomorphism if and only if $0 \to X$ is its kernel.
Dually f is an epimorphism if and only if $Y \to 0$ is its cokernel.

Proof. We will do the kernel one.

$\Longrightarrow$ Suppose f is a monomorphism.
If $g : Z \to X$ is a morphism such that $f \circ g = 0$ then by Lemma 5.0.2 $g = 0$. In particular g factors uniquely through $0 \to X$, i.e: $0 \to X$ is the kernel of f.

$\Longleftarrow$ Suppose now that $0 \to X$ is the kernel of f.
Consider a morphism $g\colon Z \to X$ such that $f \circ g = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $g = 0$, making f a monomorphism by Lemma 5.0.2.

□

Proposition 6.0.1. Let I be a small category and C an abelian category. Then the functor category $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is abelian.

Proof. By Lemma 5.0.3 we have that $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ is additive.

Furthermore since C has kernels and cokernels, which are no more than special cases of limits and colimits, by Proposition 4.3.1 $\mathsf{Fct}(\mathsf{I}, \mathsf{C})$ has kernels and cokernels and they are computed elementwise.

Finally since $\operatorname{Coim} f(i) \xrightarrow{\sim} \operatorname{Im} f(i)$ for all $i \in \mathsf{I}$ we have that $\operatorname{Coim} f \xrightarrow{\sim} \operatorname{Im} f$. □

Corollary 6.0.1. If C is abelian then the categories of complexes $\mathsf{Comp}^*(\mathsf{C})$ $(* = +, -, \mathrm{b})$ are abelian.

Proof. Direct since by Proposition 6.0.1 the category of differential objects $\mathsf{Diff}(\mathsf{C})$ is abelian. □

In what remains of the chapter we will assume that C is an abelian category.

Definition 6.0.3.

1. A complex $X \xrightarrow{f} Y \xrightarrow{g} Z$ is **exact** if $\operatorname{Im} f \xrightarrow{\sim} \ker g$. Note that this is equivalent to $\operatorname{coker} f \xrightarrow{\sim} \operatorname{Im} g$.
2. More generally, a sequence of morphisms $X^p \xrightarrow{\mathrm{d}^p} \cdots \to X^n$ with $\mathrm{d}^{i+1} \circ \mathrm{d}^i = 0$ for all $i \in \{p, \dots, n-1\}$ is **exact** if $\operatorname{Im} \mathrm{d}^i \xrightarrow{\sim} \ker \mathrm{d}^{i+1}$ for all $i \in \{p, \dots, n-1\}$.
3. A **short exact sequence** is an exact sequence of the form $0 \to X \to Y \to Z \to 0$.

Remark 6.0.1. Note that any morphism $f : X \to Y$ may be decomposed in short exact sequences

$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{Coim} f \longrightarrow 0$$

$$0 \longrightarrow \operatorname{Im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

with $\operatorname{Coim} f \cong \operatorname{Im} f$.

Prove

6.1 Exact functors

Definition 6.1.1. Let C, D be two abelian categories and
$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$
be a short exact sequence in an C.
An additive functor $F\colon \mathsf{C} \to \mathsf{D}$ is **left-exact** if
$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$
is exact in D.
Dually, it is **right-exact** if
$$F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0$$
is exact in D.
If F is both left and right exact we say that it is **exact**.

We state the following characterization of exact functors, which we will not prove.

Lemma 6.1.1 ([Sch, Lemma 5.2.2])**.** Let $F\colon \mathsf{C} \to \mathsf{D}$ be an additive functor.

1. The following are equivalent
   - **(a)** F is left exact (resp. right exact).
   - **(b)** F commutes with kernels: for any morphism $f\colon X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F f$ (resp. with cokernels).
   - **(c)** F commutes with finite limits (resp. finite colimits).
   - **(d)** For any short exact sequence $0 \to X \to Y \to Z \to 0$ in C the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in D (resp. $F(X) \to F(Y) \to F(Z) \to 0$ is exact).
2. The following are equivalent
   - **(a)** F is exact.
   - **(b)** For any exact sequence $X \to Y \to Z$ in C the sequence $F(X) \to F(Y) \to F(Z)$ is exact in D.
   - **(c)** F commutes with finite limits and finite colimits.

As a direct application of Lemma 6.1.1.

Example 6.1.1. The Hom bifunctor $\mathrm{Hom}_{\mathsf{C}}\colon \mathsf{C}^{\mathrm{op}} \times \mathsf{C} \to \mathsf{Ab}$ is left exact with respect to each of its arguments.

Example 6.1.2. Recall Theorem 4.2.1.
Right adjoints are left exact. Dually, left adjoints are right exact.

Example 6.1.3. Let I be a small category. If C admits I-indexed limits then the functor $\lim\colon \mathsf{Fct}(\mathsf{I}^{\mathrm{op}}, \mathsf{C}) \to \mathsf{C}$ is left exact. Dually if C admits I-indexed colimits then the functor $\mathrm{colim}\colon \mathsf{Fct}(\mathsf{I}, \mathsf{C}) \to \mathsf{C}$ is right exact.

Example 6.1.4. Let R be a ring and $\mathcal{I}$ an index set. The two functors
$$\prod_{i\in\mathcal{I}}\colon \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R) \qquad \bigoplus_{i\in\mathcal{I}}\colon \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$$
are exact.

Example 6.1.5. Let R be a ring and $\mathcal{I}$ a small directed category. The functor $\varinjlim_{i\in\mathcal{I}}\colon \mathsf{Fct}(\mathcal{I}, \mathsf{Mod}(R)) \to \mathsf{Mod}(R)$ is exact.

6.1.1 Diagram lemmas

Theorem 6.1.1 (Freyd-Mitchell embedding - [Wei94, Section 1.6]). Let C be a small abelian category. Then there exists a ring R (with unit, not necessarily commutative) and a fully faithful exact functor $F\colon \mathsf{C} \to \mathsf{Mod}(R)$ which embeds C as a full subcategory of $\mathsf{Mod}(R)$.

Lemma 6.1.2 (Five lemma). Consider a commutative diagram

$$\begin{array}{ccccccccc} X^0 & \xrightarrow{\alpha^0} & X^1 & \xrightarrow{\alpha^1} & X^2 & \xrightarrow{\alpha^2} & X^3 \\ \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \downarrow{\scriptstyle f^2} & & \downarrow{\scriptstyle f^3} \\ Y^0 & \xrightarrow[\beta^0]{} & Y^1 & \xrightarrow[\beta^1]{} & Y^2 & \xrightarrow[\beta^2]{} & Y^3 \end{array}$$

where the rows are exact.
If f^0 is an epimorphism and f^1, f^3 are monomorphisms then f^2 is a monomorphism. Dually if f^3 is an monomorphism and f^0, f^2 are epimorphisms then f^1 is a epimorphism.

Proof. Let $x_2 \in X^2$ such that $f^2(x_2) = 0$. We want to see that $x_2 = 0$.

We have that $(f^3 \circ \alpha^2)(x_2) = 0$ which, since f^3 is a monomorphism, implies that $\alpha^2(x_2) = 0$. Since the first row is exact there exists $x_1 \in X^1$ such that $\alpha(x_1) = x_2$.

Let $y_1 \coloneqq f^1(x_1)$. Since $(\beta^1 \circ f^1)(x_1) = 0$ and the second row is exact there exists $y_0 \in X^0$ such that $\beta^0(y_0) = f^1(x_1)$.

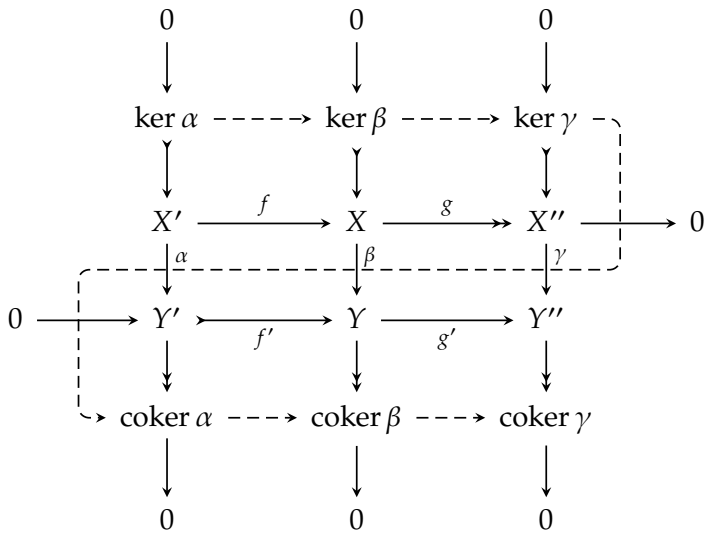
Since f^0 is an epimorphism there exists $x_0 \in X^0$ such that $y_0 = f^0(x_0)$.

Since $(f^1 \circ \alpha^0)(x_0) = f^1(x_1)$ and f^1 is a monomorphism we have $\alpha^0(x_0) = x_1$. Therefore $x_2 = \alpha^1(x_1) = 0$. □

Lemma 6.1.3 (Snake lemma). Consider a commutative diagram

$$\begin{array}{ccccccccc} & & X' & \xrightarrow{f} & X & \xrightarrow{g} & X'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & Y' & \xrightarrow[f']{} & Y & \xrightarrow[g']{} & Y'' & & \end{array}$$

where the rows are exact.
Then there exists a morphism $\delta\colon \ker \gamma \to \operatorname{coker} \alpha$ giving rise to an exact sequence



Proof. We will first check that $\ker \alpha \to \ker \beta \to \ker \gamma$ is exact. Let $x \in \ker \beta$ with $g(x) = 0$. Since the first row is exact there exists $x' \in X'$ with $f(x') = x$. Then $(f' \circ \alpha)(x') = (\beta \circ f)(x') = 0$. Since f' is a monomorphism we get $\alpha(x') = 0$ and thus $x' \in \ker \alpha$.

A dual argument shows that $\operatorname{coker} \alpha \to \operatorname{coker} \beta \to \operatorname{coker} \gamma$ is exact.

It only remains to construct δ. Let $x'' \in \ker \gamma$ and choose $x \in X$ with $g(x) = x''$. Set $y := \beta(y)$. Since $g'(y) = 0$ there exists $y' \in Y'$ such that $f'(y') = y$. Therefore we can define $\delta(x'')$ to be the image of y' in $\operatorname{coker} \alpha$. □

6.2 Complexes and cohomology

Recall that the categories $\mathsf{Comp}^*(\mathsf{C})$ are abelian for $* = +, -, \mathrm{b}, \mathrm{ub}$.

Definition 6.2.1 (Cohomology). Let

$$X^\bullet := \cdots \longrightarrow X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} X^n \xrightarrow{\mathrm{d}_X^n} X^{n+1} \longrightarrow \cdots$$

be a complex in C (i.e: $X^\bullet \in \mathsf{Comp}(\mathsf{C})$). We define

$$\begin{aligned} Z^n(X^\bullet) &:= \ker \mathrm{d}_X^n \\ B^n(X^\bullet) &= \operatorname{Im} \mathrm{d}_X^{n-1} \\ \mathrm{H}^n(X^\bullet) &:= {}^{Z^n(X^\bullet)}\!/_{B^n(X^\bullet)} = \operatorname{coker}(B^n(X^\bullet) \to Z^n(X^\bullet)) \end{aligned}$$

We call $\mathrm{H}^n(X^\bullet)$ the n-th **cohomology** of $X^\bullet$.

Lemma 6.2.1. Let C be an abelian category and $f^\bullet \colon X^\bullet \to Y^\bullet$ a morphism in $\mathsf{Comp}(\mathsf{C})$ homotopic to zero. Then $\mathrm{H}^n f^\bullet \colon \mathrm{H}^n(X^\bullet) \to \mathrm{H}^n(Y^\bullet)$ is the zero morphism for all $n \in \mathbb{Z}$.
In particular if $f^\bullet, g^\bullet : X^\bullet \to Y^\bullet$ are homotopic $\mathrm{H}^n f^\bullet = \mathrm{H}^n g^\bullet$ for all $n \in \mathbb{Z}$.

Proof. Let

$$f^n = s^{n+1} \circ \mathrm{d}_X^n + \mathrm{d}_Y^{n-1} \circ s^n$$

Then $\mathrm{d}_X^n = 0$ in $\ker \mathrm{d}_X^n$ and the image of d_Y^{n-1} is zero on ${}^{\ker \mathrm{d}_Y^n}\!/_{\operatorname{Im} \mathrm{d}_Y^{n-1}}$.

Therefore $\mathrm{H}^n f : {}^{Z^n(X^\bullet)}/_{B^n(X^\bullet)} \to {}^{Z^n(Y^\bullet)}/_{B^n(Y^\bullet)}$ is the zero morphism. □

Corollary 6.2.1. Let $X^\bullet \sim Y^\bullet$ be two homotopic complexes. Then $\mathrm{H}^n(X^\bullet) \cong \mathrm{H}^n(Y^\bullet)$.

Proof. Since $X^\bullet \sim Y^\bullet$ there exist maps $f^\bullet : X^\bullet \to Y^\bullet$ and $g^\bullet : Y^\bullet \to X^\bullet$ such that $g^\bullet \circ f^\bullet \sim \mathbb{1}_{X^\bullet}$ and $f^\bullet \circ g^\bullet \sim \mathbb{1}_{Y^\bullet}$.

Therefore by Lemma 6.2.1 for all $n \in \mathbb{Z}$ we have that

$$\mathrm{H}^n(g^\bullet \circ f^\bullet) \cong \mathrm{H}^n\, \mathbb{1}_{X^\bullet} = \mathbb{1}_{\mathrm{H}^n(X^\bullet)}$$
$$\mathrm{H}^n(f^\bullet \circ g^\bullet) \cong \mathrm{H}^n\, \mathbb{1}_{Y^\bullet} = \mathbb{1}_{\mathrm{H}^n(Y^\bullet)}$$

Furthermore, since H^n is a functor

$$\mathrm{H}^n(g^\bullet \circ f^\bullet) = \mathrm{H}^n\, g^\bullet \circ \mathrm{H}^n\, f^\bullet$$
$$\mathrm{H}^n(f^\bullet \circ g^\bullet) = \mathrm{H}^n\, f^\bullet \circ \mathrm{H}^n\, g^\bullet$$

which means that $\mathrm{H}^n\, f^\bullet$ and $\mathrm{H}^n\, g^\bullet$ are inverses of each other. □

Definition 6.2.2 (Quasi-isomorphism). Let C be an abelian category. A morphism $f^\bullet : X^\bullet \to Y^\bullet$ is a **quasi-isomorphism** (abbreviated q.is) if $\mathrm{H}^n\, f^\bullet$ is an isomorphism for all $n \in \mathbb{Z}$. In this case we say that $X^\bullet$ and $Y^\bullet$ are **quasi-isomorphic**.
In particular $X^\bullet$ is quasi-isomorphic to zero if and only if $X^\bullet$ is exact.

Definition 6.2.3 (Resolution). Let C be an abelian category and $X \in \mathsf{C}$. A **right resolution** of X is a quasi-isomorphism $X \to Y^\bullet$ with $Y^\bullet \in \mathsf{Comp}^+(\mathsf{C})$.
Dually, a **left resolution** is a quasi-isomorphism $Z^\bullet \to X$ with $Z^\bullet \in \mathsf{Comp}^-(\mathsf{C})$.

Remark 6.2.1. Let $X \to Y^\bullet$ be a right resolution of X.
This means that we have a morphism of complexes

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow{\scriptstyle \varepsilon} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & Y^0 & \longrightarrow & Y^1 & \longrightarrow & \cdots
\end{array}$$

Remark 6.2.2. We have short exact sequences

$$X^{n-1} \xrightarrow{\mathrm{d}_X^{n-1}} \ker \mathrm{d}_X^n \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow 0$$

$$0 \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \xrightarrow{\mathrm{d}_X^n} X^{n+1}$$

We can glue this into a exact sequence

$$0 \longrightarrow \mathrm{H}^n(X^\bullet) \longrightarrow \operatorname{coker} \mathrm{d}_X^{n-1} \xrightarrow{\mathrm{d}_X^n} \ker \mathrm{d}_X^{n+1} \longrightarrow \mathrm{H}^{n+1}(X^\bullet) \longrightarrow 0$$

Theorem 6.2.1. Let

$$0 \longrightarrow X^\bullet \longrightarrow Y^\bullet \longrightarrow Z^\bullet \longrightarrow 0$$

be a short exact sequence in an abelian category $\mathsf{Comp}(\mathsf{C})$.

1. For all $k \in \mathbb{Z}$ the sequence $\mathrm{H}^k(X^\bullet) \to \mathrm{H}^k(Y^\bullet) \to \mathrm{H}^k(Z^\bullet)$ is exact.
2. For all $k \in \mathbb{Z}$ there exists $\delta^k : \mathrm{H}^k(Z^\bullet) \to \mathrm{H}^{k+1}(X^\bullet)$ making the long sequence

$$\cdots \longrightarrow \mathrm{H}^{k-1}(Y^\bullet) \longrightarrow \mathrm{H}^{k-1}(Z^\bullet) \xrightarrow{\delta^{k-1}} \mathrm{H}^k(X^\bullet) \longrightarrow \mathrm{H}^k(Y^\bullet) \longrightarrow \mathrm{H}^k(Z) \xrightarrow{\delta^k} \mathrm{H}^{k+1}(X^\bullet) \longrightarrow \mathrm{H}^{k+1}(Y^\bullet) \longrightarrow \cdots$$

exact. Furthermore δ^k can be constructed to be functorial with respect to short exact sequences of $\mathsf{Comp}(\mathsf{C})$.

Proof. Consider the diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \mathrm{H}^k(X^\bullet) & & \mathrm{H}^k(Y^\bullet) & & \mathrm{H}^k(Z^\bullet) & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \operatorname{coker} \mathrm{d}_X^{k-1} & \longrightarrow & \operatorname{coker} \mathrm{d}_Y^{k-1} & \longrightarrow & \operatorname{coker} \mathrm{d}_Z^{k-1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \ker \mathrm{d}_X^{k+1} & \longrightarrow & \ker \mathrm{d}_Y^{k+1} & \longrightarrow & \ker \mathrm{d}_Z^{k+1} & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \mathrm{H}^{k+1}(X^\bullet) & & \mathrm{H}^{k+1}(Y^\bullet) & & \mathrm{H}^{k+1}(Z^\bullet) & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

where the columns are exact by Remark 6.2.2 and the rows are exact by hypothesis. Therefore the result follows from the Snake lemma. □

Chapter 7

Derived functors

7.1 Injective and projective objects

Definition 7.1.1. Let C be a category. An object $I \in \mathsf{C}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & & \\ Z & & \end{array}$$

where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \xrightarrow{k} & I \\ {\scriptstyle f}\downarrow & \nearrow_{h} & \\ Z & & \end{array}$$

Dually, an object $P \in \mathsf{C}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & & X \\ & \nearrow_{h} & \downarrow{\scriptstyle f} \\ P & \xrightarrow[k]{} & Z \end{array}$$

Notation. If C is any category we denote by $\mathcal{I}_{\mathsf{C}}$ its full subcategory made of injective objects and by $\mathcal{P}_{\mathsf{C}}$ its full subcategory of projective objects.

Proposition 7.1.1. Let C be an abelian category.

- An object $I \in \mathsf{C}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{C}}(-, I)$ is exact.
- An object $P \in \mathsf{C}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{C}}(P, -)$ is exact.

Proof. By Example 6.1.1 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in \mathcal{I}_{\mathsf{C}}$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 6.1.1 (recall that Hom is a contravariant functor), we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.

Since by Example 6.1.1 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.

Consider a morphism $\alpha \colon X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta \colon Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f \colon X \rightarrowtail Y$ in C.

We have the following short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\operatorname{coker} f, I) \longrightarrow \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.

Therefore if $\gamma \colon X \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(X, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects.

□

Definition 7.1.2. Let C be a category. We say that:

- C has **enough injectives** if for every object $X \in \mathsf{X}$ there exists an injective object $I \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $X \rightarrowtail I$.
- C has **enough projectives** if for every object $X \in \mathsf{A}$ there exists a projective object $P \in \mathcal{P}_{\mathsf{C}}$ and an epimorphism $P \twoheadrightarrow X$.

Theorem 7.1.1. Let C be an abelian category with enough injectives. Then every object $X \in \mathsf{C}$ admits a right resolution $X \to I^\bullet$ where $I^n \in \mathcal{I}_{\mathsf{C}}$ is injective for all $n \in \mathbb{Z}$.

Dually, if C has enough projectives every object $X \in \mathsf{C}$ admits a left resolution $P^\bullet \to X$ where $P^n \in \mathcal{P}_{\mathsf{C}}$ is projective for all $n \in \mathbb{Z}$.

Proof. Let $X \in \mathsf{C}$.

We want to construct an exact sequence

$$0 \longrightarrow X \longrightarrow I^0 \longrightarrow \cdots \longrightarrow I^n \longrightarrow \cdots$$

We will do it by induction over n.

For the base case, since C has enough injectives there exists an injective object $I_0 \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $X \rightarrowtail I_0$.

Suppose now that we have a exact sequence

$$0 \longrightarrow X \longrightarrow I^0 \longrightarrow \cdots \longrightarrow I^{n-1} \longrightarrow I^n$$

Since C is abelian there exists

$$\operatorname{coker}(I^{n-1} \to I^n) \colon I^n \twoheadrightarrow B^n$$

Note that

$$I^{n-1} \longrightarrow I^n \twoheadrightarrow B^n \longrightarrow 0$$

is exact. Since C has enough injectives there exists $I^{n+1} \in \mathcal{I}_{\mathsf{C}}$ and a monomorphism $B^n \rightarrowtail I^{n+1}$ Therefore

$$\begin{array}{ccccc} & & B^n & & \\ & & \uparrow & \searrow & \\ I^{n-1} & \longrightarrow & I^n & \longrightarrow & I^{n+1} \end{array}$$

is exact and induction follows. □

7.2 Derived functors

In this section C and D will be abelian categories, which we will assume that have enough injectives.

We will consider left-exact functor $F \colon \mathsf{C} \to \mathsf{D}$, but there are dual definitions and statements for right-exact functors.

Definition 7.2.1. Let C be an abelian category with enough injectives. Let $F \colon \mathsf{C} \to \mathsf{D}$ be a left-exact functor. An object $X \in \mathsf{C}$ is **F-acyclic** if for all injective resolutions $X \to I^\bullet$ of X and $n > 0$ we have

$$\mathrm{H}^n(F(I^\bullet)) = 0$$

Example 7.2.1. Injective objects are F-acyclic for all left-exact functors F.
Indeed, any injective object $I \in \mathcal{I}_{\mathsf{C}}$

Definition 7.2.2. Let C be an abelian category with enough injectives. Let $F \colon \mathsf{C} \to \mathsf{D}$ be a left-exact functor. We define the i-th **right derived functor** of F

$$\mathrm{R}^i F \colon \mathsf{C} \to \mathsf{D}$$

as, for any object $X \in \mathsf{C}$

$$\left(\mathrm{R}^i F\right)(X) \coloneqq \mathrm{H}^i(F(Q^\bullet))$$

where $Q^\bullet$ is a F-acyclic resolution of X, and for any morphism $f : X \to Y$

$$\left(\mathrm{R}^i F\right) f$$

PART III

Sheaves

Chapter 8

Presheaves and sheaves

8.1 Presheaves and sheaves

Definition 8.1.1 (Presheaf). Let X be a topological space and let C be a category. The category of C-valued presheaves over X is the functor category

$$\mathsf{PSh}_X(\mathsf{C}) := \mathsf{Fct}(\mathsf{Open}(X)^{\mathrm{op}}, \mathsf{C})$$

Less abstractly, a C-valued presheaf $\mathcal{F}$ is the collection of data:

- For each open set $U \subseteq X$ an object $\mathcal{F}(U) \in \mathsf{C}$ called *sections of $\mathcal{F}$ over U*.
- For every inclusion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set $U \subseteq X$, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclusion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms

$$\{\phi(U) \colon \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Open}(X)}$$

in C making the following diagram commute for all $V \subseteq U$

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow[\phi(V)]{} & \mathcal{G}(V) \end{array}$$

Definition 8.1.2 (Constant presheaf). Let X be a topological space. For any object $A \in \mathsf{C}$ the **constant presheaf** with value A, denoted $\underline{A}^{\mathrm{pre}}_X$ is the C-valued presheaf associated to the constant functor

$$\begin{array}{ccc} \underline{A}^{\mathrm{pre}}_X : \mathsf{Open}(X)^{\mathrm{op}} & \longrightarrow & \mathsf{C} \\ U & & A \\ {\scriptstyle V \subseteq U}\downarrow & \longmapsto & \downarrow{\scriptstyle \mathbb{1}_A} \\ V & & A \end{array}$$

Note that this defines a functor

$$\underline{-}^{\mathrm{pre}}_X : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$$

We now turn our attention to a particular class of presheaves.

Instead of working on a general category C, we will work with categories which are "sets with an algebraic structure" (for example Grp, Ab, Ring, ... but not Top).

Definition 8.1.3. A **type of algebraic structure** is a category C together with a functor $F\colon \mathsf{C} \to \mathsf{Set}$ verifying:

1. F is faithful.
2. C has limits and F commutes with them.
3. C has direct limits and F commutes with them.
4. F reflects isomorphisms.

Example 8.1.1. The following categories, together with their respective forgetful functors are types of algebraic structures:

- The category of pointed sets.
- The category of (abelian) groups.
- The category of monoids.
- The category of rings.
- The category of R-modules for a fixed ring R.
- The category of Lie algebras over a fixed field.

We state some properties of this types of categories without proof.

Lemma 8.1.1 ([Stacks, Tag 007O]). Let (C, F) be a type of algebraic structure. Then:

1. C has a terminal object 1 and $F(1) = \{*\}$.
2. C has products and $F\left(\prod X_i\right) = \prod F\left(X_i\right)$.
3. C has pullbacks and $F(X \times_Z Y) = F(X) \times_{F(Z)} F(Y)$.
4. C has equalizers and if $\mathrm{eq} \to X$ is the equalizer of $f, g\colon X \rightrightarrows Y$ then $F(\mathrm{eq}) \to F(X)$ is the equalizer of $Ff, Fg\colon F(X) \rightrightarrows F(Y)$.
5. $X \to Y$ is a monomorphism if and only if $F(X) \to F(Y)$ is injective.
6. If $Ff\colon F(X) \to F(Y)$ is surjective then f is an epimorphism.
7. Given $X_1 \to X_2 \to \cdots$, $\varinjlim X_i$ exists and $F(\varinjlim X_i) = \varinjlim F(X_i)$. And more generally for any direct limit.
8. Direct limits commute with limits.

In what follows C will be a type of algebraic structure.

Definition 8.1.4. Let X be a topological space. The category of **C-valued sheaves** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{C})$ of $\mathsf{PSh}_X(\mathsf{C})$ whose objects verify, for every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ of U

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections $(\forall i, j \in \mathcal{I} \quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j})$ there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = s_i$.

Notation. From now on, to distinguish a presheaf from a sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

Remark 8.1.1. Axioms SH.1 and SH.2 can be interpreted as the following equaliser sequence:

$$\mathscr{F}(U) \longrightarrow \prod_{i\in I} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in I\times I} \mathscr{F}(U_i \cap U_j)$$

where $\{U_i\}_{i\in I}$ is an open cover of U .
In particular taking the empty cover of $\varnothing \in \mathsf{Open}(X)$, since the empty product is terminal we have

$$\mathscr{F}(\varnothing) \longrightarrow 1 \rightrightarrows 1$$

i.e: $\mathscr{F}(\varnothing)$ is a terminal object of C

Example 8.1.2 ($\mathsf{Sh}_{\{*\}}(\mathsf{C}) \cong \mathsf{C}$). Let $\{*\}$ be the one-point topological space.

8.1.1 Stalks

Definition 8.1.5. Let X be a topological space and let $\mathcal{F}$ be a C-valued presheaf. For all points $p \in X$ we define the **stalk** of $\mathcal{F}$ at p as

$$\mathcal{F}_p \coloneqq \varinjlim_{\substack{U\ni p \\ U\in\mathsf{Open}(X)}} \mathcal{F}(U)$$

Note that this comes with maps $\mathcal{F}(U) \to \mathcal{F}_p$ for all $U \subseteq X$ open such that $p \in U$.
Since this is a direct limit following Example 3.2.5 we may describe $\mathcal{F}_p$ as

$$\mathcal{F}_p = \{(U,s) \mid p \in U, s \in \mathcal{F}(U)\}/_\sim$$

where

$$(U,s) \sim (V,t) \iff \exists W \subseteq U \cap V \mid p \in W \wedge s|_W = t|_W$$

Using this characterization we may define the stalk of a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ as

$$\begin{aligned} \phi_x \colon \mathcal{F}_x &\to \mathcal{G}_x \\ (U,s) &\mapsto (U, \phi(s)) \end{aligned}$$

Therefore we have a stalk functor

$$-_p \colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$$

By using the same definition we can extend this to a functor

$$-_p \colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$$

Remark 8.1.2. By the properties of algebraic structures, stalks commute with finite limits.

Lemma 8.1.2. Let $\mathscr{F}$ be a C-valued sheaf over a topological space X. For all open subsets $U \subseteq X$ the map $\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$ is injective.
In other words, a sheaf is completely determined by its germs.

Proof. Let $s, t \in \mathscr{F}(U)$ such that for all $s_p = t_p$ in $\mathscr{F}_p$ for all $p \in U$.

By definition of stalk there exists an open neighbourhood $V_p \subseteq U$ of p such that $s|_{V_p} = t|_{V_p}$. Since this works for all $p \in U$, we have an open cover $\{V_p\}_{p\in U}$ of U.

Therefore since $\mathscr{F}$ is a sheaf by SH.1 we have $s = t$. □

Theorem 8.1.1. Let $\phi, \psi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves where $\mathcal{F}$ is a C-valued presheaf and $\mathcal{G}$ is a C-valued sheaf. Then $\phi = \psi$ if and only if for all $p \in X$ the stalks at p are equal, i.e: $\phi_p = \psi_p$.

Proof. The direction [$\implies$] is trivial.

For the [$\impliedby$] direction suppose that $\phi_p = \psi_p$ for all $p \in X$ and consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \overset{\phi}{\underset{\psi}{\rightrightarrows}} & \mathcal{G}(U) \\ {\scriptstyle i_{\mathcal{F}}}\downarrow & & \downarrow{\scriptstyle i_{\mathcal{G}}} \\ \prod_{p\in U} \mathcal{F}_p & \overset{\prod \phi_p}{\underset{\prod \psi_p}{\rightrightarrows}} & \prod_{p\in U} \mathcal{G}_p \end{array}$$

For all $s \in \mathcal{F}(U)$ we have

$$i_{\mathcal{G}}(\phi(U)(s)) = \big(\phi_p(s_p)\big)_{p\in U} = \big(\psi_p(s_p)\big)_{p\in U} = i_{\mathcal{G}}(\psi(U)(s))$$

Since $i_{\mathcal{G}}$ is injective we have

$$\phi(U)(s) = \psi(U)(s)$$

Therefore since this holds for all $s \in \mathcal{F}(U)$ and $U \subseteq X$ open we have that $\phi = \psi$. □

Proposition 8.1.1. Let $\phi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves. Then ϕ is a isomorphism if and only if for all $p \in X$ the induced stalk map $\phi_p : \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism.

Proof.

$\implies$ Suppose that $\phi : \mathscr{F} \to \mathscr{G}$ is an isomorphism in $\mathsf{Sh}_X(\mathsf{C})$. Then there exists $\psi : \mathscr{G} \to \mathscr{F}$ such that

$$\psi \circ \phi = \mathbb{1}_{\mathscr{F}} \qquad \phi \circ \psi = \mathbb{1}_{\mathscr{G}}$$

Therefore since taking stalks is functorial for all $p \in X$ we have

$$\psi_p \circ \phi_p = \mathbb{1}_{\mathscr{F}_p} \qquad \phi_p \circ \psi_p = \mathbb{1}_{\mathscr{G}_p}$$

i.e: $\phi_p : \mathscr{F}_p \to \mathscr{G}_p$ is an isomorphism with inverse ψ_p.

$\impliedby$

□

Definition 8.1.6. Let X be a topological space.

1. A presheaf $\mathcal{F}$ is a **subpresheaf** of a presheaf $\mathcal{G}$ if for all open sets $U \subseteq X$ we have $\mathcal{F}(U) \subseteq \mathcal{G}(U)$ and the restriction maps of $\mathcal{G}$ induce the restriction maps of $\mathcal{F}$.
   If $\mathscr{F}$ and $\mathscr{G}$ are sheaves we say that $\mathscr{F}$ is a **subsheaf** of $\mathscr{G}$.
2. A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is **injective** if for all open sets $U \subseteq X$ the map $\mathcal{F}(U) \to \mathcal{G}(U)$ is injective.
3. A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is **surjective** if for all open sets $U \subseteq X$

the map $\mathcal{F}(U) \to \mathcal{G}(U)$ is surjective.

4. A morphism of sheaves $\varphi : \mathscr{F} \to \mathscr{G}$ is **injective** if for all open sets $U \subseteq X$ the map $\mathscr{F}(U) \to \mathscr{G}(U)$ is injective.
5. A morphism of sheaves $\varphi : \mathscr{F} \to \mathscr{G}$ is **surjective** if for all open sets $U \subseteq X$ and every section $s \in \mathscr{G}(U)$ there exists an open covering $U = \bigcup_{i \in \mathcal{I}} U_i$ such that $s|_{U_i}$ is in the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$ for all i.

8.2 Sheafification

We have established the definition of presheaves and sheaves. It is clear from the definitions that every sheaf is a presheaf. This defines a "forgetful" functor:

$$\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$$

We want to associate to each presheaf a sheaf.

Definition 8.2.1 (Sheafification). Let X be a topological space and C a category. The **sheafification** (or associated sheaf) functor, denoted by $-^{\#} : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$, is the left adjoint to the inclusion functor $\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$.
That is, for every presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and every sheaf $\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection:

$$\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathcal{F}^{\#}, \mathscr{G}) \cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota\mathscr{G})$$

The unit of this adjunction gives a canonical morphism $\theta : \mathcal{F} \mapsto \mathcal{F}^{\#}$ in $\mathsf{PSh}_X(\mathsf{C})$ which verifies that for any presheaf morphism $\phi : \mathcal{F} \to \mathscr{G}$ where $\mathcal{F}$ is a presheaf and $\mathscr{G}$ is a sheaf there exists a unique morphism $\psi : \mathcal{F}^{\#} \to \mathscr{G}$ such that $\psi \circ \theta = \phi$.

Construction (Godement construction of sheafification). Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$.

For every open set $U \subseteq X$, let

$$\Lambda(\mathcal{F})(U) := \prod_{p \in U} \mathcal{F}_p$$

Considering for $V \subseteq U$ the morphism

$$\begin{aligned} \mathrm{res}_{U,V} : \Lambda(\mathcal{F})(U) &\to \Lambda(\mathcal{F})(V) \\ \prod_{p \in U} \mathcal{F}_p &\mapsto \prod_{p \in V} \mathcal{F}_p \end{aligned}$$

as the restriction maps, we see that $\Lambda(\mathcal{F})$ is a C-valued presheaf.

Let's check that $\Lambda(\mathcal{F})$ verifies the sheaf axioms. To do this, consider an open set $U \subseteq X$.

SH.1) Let $s, t \in \mathcal{F}(U)$ be two sections such that there exists an open covering $\{U_i\}_{i \in \mathcal{I}}$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = t|_{U_i}$.
Note that for any $x \in U$ since there exists $j \in \mathcal{I}$ such that $x \in U_j$ we have that $s_x = t_x$. Therefore, since this holds for all $x \in U$, we have that $s = t$.

SH.2) Let $\{U_i\}_{i\in\mathcal{I}}$ be an open covering of U and let $\{s_i \mid s_i \in \Lambda(\mathcal{F})(U_i)\}_{i\in\mathcal{I}}$ be a family of sections such that for any $i, j \in \mathcal{I}$ we have $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$.
Define a section $s \in \Lambda(\mathcal{F})(U)$ by choosing, for each $x \in U$, an index $k \in \mathcal{I}$ such that $x \in U_k$ and setting $s_x = (s_k)_x$. Note that this is well-defined by the compatibility condition of the family. Furthermore, for all $i \in \mathcal{I}$ by construction we have that $s|_{U_i} = s_i$.

Note that there is a presheaf morphism $\theta\colon \mathcal{F} \to \Lambda(\mathcal{F})$ defined for any $s \in \mathcal{F}(U)$ by sending s to its germ at every point:

$$\theta(U)(s) = (s_p)_{p\in U} \in \prod_{p\in U} \mathcal{F}_p$$

where s_p denotes the image of s in the stalk $\mathcal{F}_p$.

We define the sheafification $\mathcal{F}^\#$ of $\mathcal{F}$ as the subpresheaf of $\Lambda(\mathcal{F})$ consisting of sections that are *locally compatible* with $\mathcal{F}$. Explicitly, for every open set $U \subseteq X$ an element $\sigma = (\sigma_p)_{p\in U} \in \Lambda(\mathcal{F})(U)$ belongs to $\mathcal{F}^\#(U)$ if and only if:

> For every $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and a section $t \in \mathcal{F}(V)$ such that for all $q \in V$, $\sigma_q = t_q$ (as elements of $\mathcal{F}_q$).

We need to check that our definition of $\mathcal{F}^\#$ verifies the universal propery of sheafification. To do this, consider a sheaf $\mathcal{G}$ and a morphism of presheaves $\phi\colon \mathcal{F} \to \mathcal{G}$. We wish to show that there exists a unique sheaf morphism $\psi\colon \mathcal{F}^\# \to \mathcal{G}$ such that $\phi \circ \theta = \psi$.

Let's check existence first by defining ψ explicitly. Let $U \subseteq X$ be an open set and $s \in \mathcal{F}^\#(U)$. By definition there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U and sections $t_i \in \mathcal{F}(V_i)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = \theta(t_i)$.

We want ψ to verify $\psi(s)|_{U_i} = \psi(s|_{U_i}) = \psi(\theta_{t_i})$. This forces us to define $\psi(\theta(t_i)) = \phi(t_i) \in \mathcal{G}(U_i)$.

Let's see that $\{\phi(t_i)\}_{i\in\mathcal{I}}$ can be glued in $\mathcal{G}$. Note that for all $y \in U_i \cap U_j$ we have $(t_i)_y = (t_j)_y$. Therefore using the induced stalk morphism $\phi_y\colon \mathcal{F}_y \to \mathcal{G}_y$ we have $\phi_y((t_i)_y) = \phi_y((t_j)_y)$. Therefore $(\phi(t_i))_y = (\phi(t_j))_y$ for all $y \in U_i \cap U_j$, meaning that $\phi(t_i)|_{U_i\cap U_j} = \phi(t_j)|_{U_i\cap U_j}$.

Applying SH.2 to this family we get that there is a unique section $t \in \mathcal{G}(U)$ such that $\forall i \in \mathcal{I} \quad t|_{U_i} = \phi(t_i)$. Therefore we can define $\psi_U(s) = t$, which verifies the universal property by construction.

To check uniqueness, since sheaf morphisms are determined by their induced maps on stalks and for all points $x \in X$ we have $\theta_x = \mathbb{1}$, ψ verifies $\forall x \in X \quad \psi_x = \phi_x$ making it unique.

Lemma 8.2.1. Let $\mathcal{F}$ be a sheaf. Then $(\iota\mathcal{F})^\# \cong \mathcal{F}$, i.e: $-^\# \circ \iota \cong \mathbb{1}_{\mathsf{Sh}_X(\mathsf{C})}$.

Proof. Consider the identity map $\mathbb{1} : \iota\mathcal{F} \to \mathcal{F}$. Since $\iota\mathcal{F}$ is a presheaf and $\mathcal{F}$ is a sheaf by definition of sheafification there exists a unique morphism $\psi : (\iota\mathcal{F})^\# \to \mathcal{F}$ such that $\psi \circ \theta = \mathbb{1}$. Note that

$$(\theta \circ \psi) \circ \theta = \theta \circ (\psi \circ \theta) = \theta$$

Applying now the definition of sheafification to $\theta : \mathscr{F} \to (\iota\mathscr{F})^{\#}$ we get a unique map $h : (\iota\mathscr{F})^{\#} \to (\iota\mathscr{F})^{\#}$ such that $h \circ \theta = \theta$. Therefore since $\mathbb{1} \circ \theta = \theta$ and $(\theta \circ \psi) \circ \theta = \theta$ we have that $\theta \circ \psi = \mathbb{1}$, which means that θ is an isomorphism. □

Lemma 8.2.2. Let X be a topological space and $\mathcal{F}$ a C-valued presheaf over X. Then for all $p \in X$ there is bijection of stalks

$$\mathcal{F}_p \cong (\mathcal{F}^{\#})_p$$

Proof. Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and $p \in X$. For all $A \in \mathsf{C}$ we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{C}}\left((\mathcal{F}^{\#})_p, A\right) &\cong \mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}\left(\mathcal{F}^{\#}, \mathrm{sk}_X(A)\right) && \left(-_p \dashv \mathrm{sk}_X\right)\\
&\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\mathcal{F}, \iota\,\mathrm{sk}_X(A)\right) && \left(-^{\#} \dashv \iota\right)\\
&\cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}\left(\mathcal{F}, \mathrm{sk}_X^{\mathrm{pre}}(A)\right) && \left(\iota\,\mathrm{sk}_X = \mathrm{sk}_X^{\mathrm{pre}}\right)\\
&\cong \mathrm{Hom}_{\mathsf{C}}\left(\mathcal{F}_p, A\right) && \left(-_p \dashv \mathrm{sk}_X^{\mathrm{pre}}\right)
\end{aligned}$$

Therefore by the Yoneda lemma $(\mathcal{F}^{\#})_p \cong \mathcal{F}_p$. □

8.3 Sheaves in an abelian category

In this section, X will be a fixed topological space and C will be an abelian category for which sheafification exists.

Note that by Proposition 6.0.1 $\mathsf{PSh}_X(\mathsf{C})$ is an abelian category.

We are going to see that $\mathsf{Sh}_X(\mathsf{C})$ is an abelian category.

Consider the following category-theoretic auxiliary lemma.

Lemma 8.3.1. Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two functors. Suppose that

1. A and B are additive categories, a, b are additive functors and there is an adjunction $b \dashv a$.
2. B is an abelian category and b is left-exact.
3. $b \circ a \cong \mathbb{1}_{\mathsf{A}}$.

Then A is abelian.

Proof. Since B is an abelian by Lemma 6.0.1 all finite limits and colimits exist en B.

Since b is left-adjoint by Example 6.1.2 it is right-exact and therefore, exact. Analogously we have that a is left-exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. Since B is abelian there exist $\ker_{\mathsf{B}} \varphi$ and $\mathrm{coker}_{\mathsf{B}} \varphi$.

Since exact functors preserve kernels and cokernels we have that

$$b(\ker_{\mathsf{B}} \varphi) \cong \ker_{\mathsf{A}}(b\varphi) \qquad b(\mathrm{coker}_{\mathsf{B}}\varphi) \cong \mathrm{coker}_{\mathsf{A}}(b\varphi)$$

Therefore we have that every morphism of the form $b\varphi$ in A admits a kernel and a cokernel.

Note that, since $b \circ a \cong \mathbb{1}_{\mathsf{A}}$, every morphism ψ in A is of the form $\psi = b\varphi$. Thus all kernels and cokernels exist in A and

$$\ker_{\mathsf{A}} \psi \cong b \ker_{\mathsf{B}}(a\psi) \qquad \operatorname{coker}_{\mathsf{A}} \psi \cong b \operatorname{coker}_{\mathsf{B}}(a\psi) \tag{8.3.1}$$

It only remains to show that $\operatorname{Coim}_{\mathsf{A}}\psi \cong \operatorname{Im}_{\mathsf{A}}\psi$. Since A has all finite limits and colimits a is left exact.

By Lemma 6.1.1 we have that a transforms kernels into kernels. Therefore

$$\begin{aligned}
\operatorname{coim}_{\mathsf{A}}\psi &= \operatorname{coker}_{\mathsf{A}}\left(\ker_{\mathsf{A}} \psi \to A_1\right) && \text{(definition)}\\
&\cong b\operatorname{coker}_{\mathsf{B}}\left(a(\ker_{\mathsf{A}} \psi \to A_1)\right) && \text{(8.3.1)}\\
&\cong b\operatorname{coker}_{\mathsf{B}}\left(\ker_{\mathsf{B}}(a\psi) \to a(A_1)\right) && (a \text{ preserves kernels})\\
&= b\operatorname{Coim}_{\mathsf{B}}(a\psi) && \text{(definition)}\\
&\cong b\operatorname{Im}_{\mathsf{B}}(a\psi) && (\mathsf{B} \text{ is abelian})\\
&= b(\ker_{\mathsf{B}}\left(a(A_2) \to \operatorname{coker}_{\mathsf{B}}(a\psi)\right)) && \text{(definition)}\\
&\cong \ker_{\mathsf{A}}\left((b \circ a)(A_2) \to b(\operatorname{coker}_{\mathsf{B}}(a\psi))\right) && (b \text{ preserves kernels})\\
&\cong \ker_{\mathsf{A}}\left(A_2 \to b(\operatorname{coker}_{\mathsf{B}}(a\psi))\right) && b \circ a \cong \mathbb{1}_{\mathsf{A}}\\
&\cong \ker_{\mathsf{A}}\left(A_2 \to \operatorname{coker}_{\mathsf{A}}\psi\right) && \text{(8.3.1)}\\
&= \operatorname{Im}_{\mathsf{A}}\psi && \text{(definition)}
\end{aligned}$$

□

Lemma 8.3.2. $\mathsf{Sh}_X(\mathsf{C})$ is an additive category and the forgetful functor $\iota\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is additive.

Proof. Let's check first that $\mathsf{Sh}_X(\mathsf{C})$ satisfies additive category axioms. Recall that Ad.3 is equivalent to Ad.4 by Lemma 5.0.1.

We will use the characterization of sheaves in terms of equalisers.

Ad.1 Since $\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F},\mathscr{G}) = \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{F},\mathscr{G})$ and $\mathsf{PSh}_X(\mathsf{C})$ is additive, $\operatorname{Hom}_{\mathsf{Sh}_X}$ admits the structure of an abelian group.

Ad.2 Note that the zero object in $\mathsf{PSh}_X(\mathsf{C})$ is given by the constant presheaf $\underline{0}^{\mathrm{pre}}_X$. Note that the sequence

$$\underline{0}^{\mathrm{pre}}_X(U) = 0 \longrightarrow \prod_{i\in I} \underline{0}^{\mathrm{pre}}_X(U_i) = 0 \rightrightarrows \prod_{(i,j)\in I\times I} \underline{0}^{\mathrm{pre}}_X(U_i \cap U_j) = 0$$

is trivially an equaliser for all open covers $U = \bigcup_{i\in I} U_i$. Therefore $\underline{0}^{\mathrm{pre}}_X$ is a sheaf and thus $\mathsf{Sh}_X(\mathsf{C})$ admits a zero object.

Ad.4 Let $\mathscr{F},\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$. We define their product elementwise as

$$(\mathscr{F} \times \mathscr{G})(U) = \mathscr{F}(U) \times \mathscr{G}(U)$$

Since this coincides with the product in $\mathsf{PSh}_X(\mathsf{C})$ it is a presheaf. Furthermore for any open cover $U = \bigcup_{i\in I} U_i$ we have that

$$\begin{aligned}
(\mathscr{F} \times \mathscr{G})(U) = \mathscr{F}(U) \times \mathscr{G}(U) \to \prod_{i\in I}(\mathscr{F} \times \mathscr{G})(U_i) &= \prod_{i\in I}(\mathscr{F}(U_i)) \times \prod_{i\in I}(\mathscr{G}(U_i))\\
\rightrightarrows \prod_{(i,j)\in I\times I}(\mathscr{F} \times \mathscr{G})(U_i \cap U_j) &= \prod_{i\in I}\left(\mathscr{F}(U_i \cap U_j)\right) \times \prod_{i\in I}\left(\mathscr{G}(U_i \cap U_j)\right)
\end{aligned}$$

is an equaliser since it can be seen as the "product" of two equalisers. Therefore $\mathscr{F} \times \mathscr{G}$ is a sheaf.

By the construction of the zero object and products in $\mathsf{Sh}_X(\mathsf{C})$ we have that ι preserves both. Therefore by Proposition 5.0.1 ι is additive. □

Proposition 8.3.1. Let C be an abelian category. The sheafification functor $-^{\#}\colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$ is exact.

Proof. Since $-^{\#}$ is left adjoint by Example 6.1.2 it is right exact.

To see that it is left exact we will use Lemma 6.1.1. In particular we are going to see that sheafification commutes with finite products.

Let $\mathcal{F}\colon \mathsf{I} \to \mathsf{PSh}_X(\mathsf{C})$ be a diagram in $\mathsf{PSh}_X(\mathsf{C})$ with I finite.

We need to see that

$$\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#} \cong \lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}$$

Note that since $\iota\colon \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is right adjoint it preserves limits. Therefore limits of sheaves are computed exactly as limits of presheaves.

By definition of limit we have a morphism

$$\phi\colon \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#} \to \lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}$$

To see that this is an isomorphism we will see that the induced stalk maps ϕ_p are isomorphisms for all $p \in X$, i.e: we want to show that

$$\left(\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}\right)_p$$

On the one hand since stalks commute with sheafification we have

$$\left(\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)_p$$

Furthermore since stalks commute with finite limits

$$\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i)_p$$

On the other hand since stalks commute with finite limits

$$\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i^{\#})_p$$

Furthermore since stalks commute with sheafification we have

$$\lim_{i\in\mathsf{I}} \left((\mathcal{F}_i)^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i)_p$$

Therefore

$$\left(\left(\lim_{i\in\mathsf{I}} \mathcal{F}_i\right)^{\#}\right)_p \cong \lim_{i\in\mathsf{I}} (\mathcal{F}_i)_p \cong \left(\lim_{i\in\mathsf{I}} \mathcal{F}_i^{\#}\right)_p$$

□

Theorem 8.3.1. Let $\varphi\colon \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{C})$.

1. The category $\mathsf{Sh}_X(\mathsf{C})$ is abelian.
2. The kernel of φ is the same as the kernel of φ in $\mathsf{PSh}_X(\mathsf{Ab})$.
3. φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.
4. The cokernel of φ is the sheafification of the cokernel of φ in $\mathsf{PSh}_X(\mathsf{Ab})$.
5. φ is surjective if and only if φ is surjective as a map of sheaves.
6. A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$ if and only if for all $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact.

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 4, but these come directly from Lemma 8.3.1 with

$$\mathsf{A} \coloneqq \mathsf{Sh}_X(\mathsf{C}) \quad \mathsf{B} \coloneqq \mathsf{PSh}_X(\mathsf{B})$$
$$a \coloneqq \iota\colon \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C}) \text{ forgetful} \quad b \coloneqq -^{\#}\colon \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C}) \text{ sheafification}$$

We can indeed apply this lemma since

1. $\mathsf{Sh}_X(\mathsf{C})$ is additive by Lemma 8.3.2 and $\mathsf{PSh}_X(\mathsf{C})$ is additive by Lemma 5.0.3, ι is additive by Lemma 8.3.2 and $-^{\#}$ is additive by Proposition 5.0.2, and the adjunction $-^{\#} \dashv \iota$ is given by definition of sheafification.
2. $\mathsf{PSh}_X(\mathsf{C})$ is abelian by Proposition 6.0.1 and $-^{\#}$ is left exact by Proposition 8.3.1.
3. $-^{\#} \circ \iota = \mathbb{1}_{\mathsf{Sh}_X(\mathsf{C})}$ comes from Lemma 8.2.1.

□

PART IV

Sheaf cohomology

Chapter 9

Cohomology of sheaves

Let $(X, \mathscr{O}_X)$ be a ringed space.

We want to define the cohomology of X with coefficients in a sheaf of $\mathscr{O}_X$-modules as the right derived functor of the global sections functor

$$\Gamma(X, -) : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$$

To do this, we need to check the following:

1. $\mathsf{Mod}(\mathscr{O}_X)$ and $\mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$ are abelian categories.
2. The functor $\Gamma(X, -)$ is left exact.
3. $\mathsf{Mod}(\mathscr{O})_X$ has enough injectives.

In **??** we have proved that the category of sheaves of $\mathscr{O}_X$-modules is abelian. Furthermore, since $\Gamma(X, \mathscr{O}_X)$ is a ring, $\mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$ is also abelian.

Recall that by **??** $\Gamma(X, -)$ is right adjoint. Therefore by **??** $\Gamma(X, -)$ is left exact.

It only remains to show that $\mathsf{Mod}(\mathscr{O}_X)$ has enough injectives.

Lemma 9.0.1. Let $(X, \mathscr{O}_X)$ be a ringed space. The category $\mathsf{Mod}(\mathscr{O}_X)$ of sheaves of $\mathscr{O}_X$-modules over X has enough injectives.

Proof. We will use the following claim

> The category $\mathsf{Mod}(R)$ of modules over a ring R has enough injectives.

For a proof, see [Alu09, Corollary VIII.6.4].

Let $\mathscr{F}$ be an $\mathscr{O}_X$-module. Let $\mathscr{F}_p \in \mathsf{Mod}(\mathscr{O}_{X,p})$ be that stalk of $\mathscr{F}$ at a point $p \in X$.

Since $\mathsf{Mod}(\mathscr{O}_{X,p})$ has enough injectives we can choose an object I_p and a monomorphism

$$i_p : \mathscr{F}_p \rightarrowtail I_p$$

Consider the sheaf

$$\mathscr{I} := \bigoplus_{p \in X} \iota_{p,*} \underline{I_p}$$

where $\iota_{p,*}I_p$ is the skyscraper sheaf. More explicitly, for every open set $U \subseteq X$ the sheaf $\mathscr{I}$ is defined as

$$\mathscr{I}(U) = \bigoplus_{p \in U} I_p$$

Recall that by **??** the skyscraper functor is left-adjoint to the stalk functor, which is exact. Therefore by **??** $\mathscr{I}$ is an injective object.

Furthermore, there exists a monomorphism $\phi : \mathscr{F} \to \mathscr{I}$ defined as, for any open set $U \subseteq X$

$$\begin{aligned} \phi(U) : \mathscr{F}(U) &\to \mathscr{I}(U) \\ s &\mapsto \bigoplus_{p \in U} j_p(s_p) \end{aligned}$$

□

Having seen this, we now can finally define sheaf cohomology.

Definition 9.0.1 (Sheaf Cohomology). Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules over a topological space X. The i-th **cohomology** of X with coefficients in $\mathscr{F}$ as the i-th right derived functor of the global sections functor:

$$\mathrm{H}^i(X, \mathscr{F}) \coloneqq R^i\Gamma(X, \mathscr{F})$$

Therefore, given a sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ to compute its cohomology we *only* need to construct an injective (or $\Gamma(X, -)$ acyclic by **??**) resolution of $\mathscr{F}$

$$0 \longrightarrow \mathscr{F}^{(0)} \longrightarrow \mathscr{F}^{(1)} \longrightarrow \cdots$$

and compute the cohomology of the $\Gamma(X, \mathscr{O}_X)$-modules

$$0 \longrightarrow \Gamma(X, \mathscr{F}^{(0)}) \longrightarrow \Gamma(X, \mathscr{F}^{(1)}) \longrightarrow \cdots$$

Since injective resolutions are usually hard to find (note how Lemma 9.0.1 doesn't give an *easy* construction) we will mostly work with $\Gamma(X, -)$ acyclic resolutions.

9.1 The Godement resolution

Definition 9.1.1 (Flasque sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **flasque** (or **flabby**) if for any inclussion of open subsets $V \subseteq U$ the restriction map $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ is surjective.

Proposition 9.1.1. Flasque sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a flasque sheaf. Since $\mathsf{Mod}(\mathscr{O}_X)$ has enough injectives we have a short exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{I}$ is an injective object.

We will first show that, for any open subset $U \subseteq X$, the map $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ is surjective. Let $s \in \mathscr{G}(U)$. Let $V, W \subseteq U$ be two open subsets and $\tau_V \in \mathscr{I}(V), \tau_W \in \mathscr{I}(W)$ be two sections lifting s to $\mathscr{I}$. Consider

$$\tau_V|_{V\cap U} - \tau_W|_{V\cap W} \in \mathscr{F}(V \cap W)$$

Since $\mathscr{F}$ is flasque there exists a section $\chi_V \in \mathscr{F}(V)$ such that $\chi_V|_{V\cap W} = \tau_V - \tau_W$. Therefore if $\tau'_V = \tau_V - \chi_V$ we have $\tau'_V|_{V\cap W} = \tau_W|_{V\cap W}$, i.e: there exists a section $\tau \in \mathscr{I}(V \cup W)$ such that $\tau|_V = \tau'|_V$ and $\tau'|_W = \tau|_W$. Clearly τ is sent to $\sigma|_{V\cap W}$.

Consider now a pair (W, τ) which is maximal (for the obvious order relation), where W is an open subset of U and τ is a section of $\mathscr{I}$ lifting σ to W. Since σ lifts locally on $\mathscr{I}$ and using the previous result we see that $W = U$.

But then the long exact sequence in cohomology of our sequence and the fact that for $i > 0 \quad \mathrm{H}^i(X, \mathscr{I}) = 0$ imply that $\mathrm{H}^1(X, \mathscr{F}) = 0$ and

$$\forall k - 1 \geq 1 \quad \mathrm{H}^k(X, \mathscr{F}) = \mathrm{H}^{k-1}(X, \mathscr{G})$$

Furthermore surjectivity of the maps $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ implies that $\mathscr{G}$ is flasque.

It follows by induction that

$$\forall k > 0 \quad \mathrm{H}^k(X, \mathscr{F}) = 0$$

□

Using flasque sheaves we can construct a canonical (and functorial) resolution of a sheaf of $\mathscr{O}_X$-modules to compute its cohomology.

Construction (Godement resolution)**.** Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. The **Godement resolution** of $\mathscr{F}$ is constructed by considering the inclussion of $\mathscr{F}$ into the *Godement sheaf* $\mathscr{F}_{\mathrm{God}}$

$$U \mapsto \bigoplus_{p\in U} \mathscr{F}_p =: \mathscr{F}_{\mathrm{God}}(U)$$

Note that $\mathscr{F}_{\mathrm{God}}$ is flasque.

We then inject the quotient $\mathscr{F}_{\mathrm{God}} / \mathscr{F}$ into $\left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}$ and so on.

Example 9.1.1. Let's try to compute sheaf cohomology using the Godement resolution.
Let $X = \{p, q\}$ with the *Sierpiński* topology $\mathcal{T} = \{\varnothing, \{p\}, X\}$. Consider the constant sheaf supported at the integers $\mathscr{F} = \underline{\mathbb{Z}}$.
To compute the cohomology of $\mathscr{F}$, we are going to construct the complex of global sections:

$$0 \longrightarrow \Gamma(X, \mathscr{F}_{\mathrm{God}}) \xrightarrow{d^0} \Gamma\left(X, \left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}\right) \xrightarrow{d^1} \cdots$$

Note that the stalks of $\mathscr{F}$ are:

$$\mathscr{F}_p = \mathscr{F}(\{p\}) = \mathbb{Z}, \qquad \mathscr{F}_q = \mathscr{F}(X) = \mathbb{Z}.$$

Consider the Godement sheaf associated to $\mathscr{F}$:

$$\mathscr{F}_{\text{God}}(U) = \bigoplus_{x \in U} \mathscr{F}_x.$$

We have that:

$$\Gamma(X, \mathscr{F}_{\text{God}}) = \mathscr{F}_{\text{God}}(X) = \mathscr{F}_p \oplus \mathscr{F}_q = \mathbb{Z} \oplus \mathbb{Z}.$$

This means that we have a map d^{-1} as follows:

$$d^{-1} : \underbrace{\Gamma(X, \mathscr{F})}_{\mathbb{Z}} \to \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}}, \quad d^{-1}(k) = (k, k).$$

Let $\mathscr{Q}^1 = {}^{\mathscr{F}_{\text{God}}}\big/_{\mathscr{F}}$ be the cokernel sheaf. If we compute its stalks, we have that:

$$(\mathscr{Q}^1)_p = {}^{\mathscr{F}_{\text{God},p}}\big/_{\mathscr{F}_p} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$(\mathscr{Q}^1)_q = {}^{\mathscr{F}_{\text{God},q}}\big/_{\mathscr{F}_q} = {}^{\mathbb{Z}\oplus\mathbb{Z}}\big/_{\langle (k,k) \rangle} \cong \mathbb{Z}.$$

Therefore, the global sections of the Godement sheaf of $\mathscr{Q}^1$ are:

$$\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right) = (\mathscr{Q}^1)_p \oplus (\mathscr{Q}^1)_q = 0 \oplus \mathbb{Z} = \mathbb{Z}.$$

This produces a differential map:

$$d^0 : \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}} \to \underbrace{\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right)}_{\mathbb{Z}}, \quad d^0(m, n) = m - n.$$

Let $\mathscr{Q}^2$ be the cokernel of the map $\mathscr{F}_{\text{God}} \to (\mathscr{Q}^1)_{\text{God}}$. If we compute its stalks, we have:

$$(\mathscr{Q}^2)_p = {}^{0}\big/_{0} = 0,$$
$$(\mathscr{Q}^2)_q = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0.$$

Therefore, we have that:

$$\Gamma\left(X, (\mathscr{Q}^2)_{\text{God}}\right) = 0 \oplus 0 = 0.$$

This means that our resolution of global sections is:

$$0 \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \xrightarrow[(m,n)\mapsto m-n]{d^0} \mathbb{Z} \longrightarrow 0$$

Therefore, the cohomology groups of X with coefficients in $\underline{\mathbb{Z}}$ are:

$$\mathrm{H}^0(X, \underline{\mathbb{Z}}) = \ker(d^0) = \{(m, n) \in \mathbb{Z} \oplus \mathbb{Z} \mid m - n = 0\} \cong \mathbb{Z},$$
$$\mathrm{H}^1(X, \underline{\mathbb{Z}}) = \operatorname{coker}(d^0) = {}^{\mathbb{Z}}\big/_{\operatorname{im}(d^0)} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$\mathrm{H}^i(X, \underline{\mathbb{Z}}) = 0 \quad \forall i > 1.$$

Notice that if the Godement resolution makes computing sheaf cohomology difficult on a *simple* space, it renders the task nearly impossible on more *complicated* spaces.

To address this, the upcoming chapters will explore how we can select different,

more efficient resolutions to compute cohomology on sufficiently nice topological spaces.

Chapter 10

Cohomology and homotopy

Our goal in this chapter will be to prove that sheaf cohomology of locally constant sheaves are a homotopy invariant.

We will denote $I := [0,1]$ the unit interval.

10.1 Sheaves on the interval $[0,1]$

Lemma 10.1.1. Let $\mathscr{F} \in \mathsf{Mod}(\underline{R}_X)$. Then

1. For $j > 1$ we have $\mathrm{H}^j(I;\mathscr{F}) = 0$.
2. If $\mathscr{F}(I) \to \mathscr{F}_t$ is an epimorphism for all $t \in I$ then $\mathrm{H}^1(I;\mathscr{F}) = 0$.

Proof. Let $j \geq 1$ and $s \in \mathrm{H}^j(I;\mathscr{F})$.

For $0 \leq t_1 < t_2 \leq 1$ consider the morphism

$$f_{t_1,t_2} : \mathrm{H}^j(I;\mathscr{F}) \to \mathrm{H}^j([t_1,t_2],\mathscr{F})$$

and let

$$J := \{t \in [0,1] \mid f_{0,t}(s) = 0\}$$

Since $\mathrm{H}^j(\{0\},\mathscr{F}) = 0$ we have that $0 \in J$. Furthermore since $f_{0,t}(s) = 0$ implies $f_{0,t'}(s) = 0$ for $t' \in [0,t]$, J is an interval and thus connected.

We want to see that J is both open and closed.

Open By ... we have that

$$\mathrm{H}^j([0,t_0];\mathscr{F}) \cong \varinjlim_{t>t_0} \mathrm{H}^j([0,t];\mathscr{F})$$

Therefore J is open.

Closed For $t \in [0,t_0]$ consider the Mayer-Vietoris sequence

$$\cdots \longrightarrow \mathrm{H}^j([0,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^j([0,t];\mathscr{F}) \oplus \mathrm{H}^j([t,t_0];\mathscr{F}) \longrightarrow \mathrm{H}^j(\{t\},\mathscr{F}) \longrightarrow \cdots$$

For $j > 1$ or for $j = 1$ and assuming that $\mathrm{H}^0(I;\mathscr{F}) \twoheadrightarrow \mathrm{H}^0(\{t\},\mathscr{F})$ is surjective we get

$$\mathrm{H}^j([0,t_0],\mathscr{F}) \cong \mathrm{H}^j([0,t];\mathscr{F}) \oplus \mathrm{H}^j([t,t_0];\mathscr{F})$$

Let $t_0 = \sup\{t \mid t \in J\}$. Then $f_{0,t}(s) = 0$ for all $t < t_0$. On the other hand by ...

$$\varinjlim_{t<t_0} \mathrm{H}^j([t, t_0], \mathscr{F}) \cong \mathrm{H}^j(\{t_0\}; \mathscr{F}) = 0$$

Therefore there exists $t < t_0$ with $f_{t,t_0}(s) = 0$ which implies $f_{0,t_0}(s) = 0$ meaning that $t_0 \in J$.

□

10.2 Invariance by homotopy

Definition 10.2.1. Let X, Y be two topological spaces

- Two continuous maps $f_0, f_1 : X \rightrightarrows Y$ are **homotopic** if there exists a continuous map $h : I \times X \to Y$ such that, for all $x \in X$

$$f_0(x) = h(0, x) \quad f_1(x) = h(1, x)$$

  We denote this by $f_0 \simeq f_1$.
- A continuous map $f : X \to Y$ is a **homotopy equivalence** if there exists a continuous map $g : Y \to X$ such that

$$g \circ f \simeq \mathrm{id}_X \quad f \circ g = \mathrm{id}_Y$$

  In this case we say that X and Y are **homotopic** and denote it by $X \simeq Y$.
- A topological space Z is **contractible** if $Z \simeq \{*\}$.

Theorem 10.2.1 (Invariance by homotopy). Let $f_0, f_1 : X \rightrightarrows Y$ be two homotopic maps and let $\mathscr{G}$ be a locally constant sheaf of R-modules over Y.

Chapter 11

de Rham Cohomology

Definition 11.0.1 (Partition of unity). Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. A **partition of unity** of $\mathscr{F}$ subordinate to a locally finite open covering $\mathcal{U} = \{U_i\}_{i\in\mathcal{I}}$ is a collection of endomorphisms of $\mathscr{O}_X$-modules

$$\{f_i : \mathscr{F} \to \mathscr{F}\}_{i\in\mathcal{I}}$$

satisfying

1. For all $i \in \mathcal{I}$
$$\operatorname{supp} f_i := \overline{\{x \in X \mid (f_i)_x \neq 0\}} \subset U_i$$
2. $$\sum_{i\in\mathcal{I}} f_i = \mathbb{1}_{\mathscr{F}}$$

   Note that this sum is well defined since for any point $x \in X$ there is an open neighbourhood where all but finitely many of the maps f_i are the zero maps, i.e: the sum is locally finite.

Definition 11.0.2 (Fine sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **fine** if for every open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to this covering.

Proposition 11.0.1. Fine sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a fine sheaf. Using the Godement resolution we have that $\mathscr{F}$ admits a flasque resolution $\mathscr{F}[0] \to \mathscr{I}^\bullet$.

Using Proposition 9.1.1 by **??** we have that

$$\mathrm{H}^k(X, \mathscr{F}) = \ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right) \Big/ \operatorname{im}\left(\Gamma(X, \mathscr{I}^{k-1}) \to \Gamma(X, \mathscr{I}^k)\right)$$

Let $\alpha \in \ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right)$. Since $\mathscr{I}^\bullet$ is locally exact in degree greater than 0 we have that α comes from $\mathscr{I}^{k-1}$, i.e: there exists an open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X such that $\alpha|_{U_i} = d\beta_i$ with $\beta_i \in \mathscr{I}^{k-1}(U_i)$.

Consider a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to $\mathscr{U}$ and let

$$\beta := \sum_{i\in\mathcal{I}} f_i\beta_i$$

where the sum is locally finite. Note that $f_i\beta_i$ is a section of $\mathscr{I}^{k-1}(X)$ with values in $\mathscr{I}_p^{k-1}(X)$:

$$(f_i\beta_i)_p = \begin{cases} 0 & p \notin U_i \\ f_i\beta_i & p \in U_i \end{cases}$$

Since f_i has support in U_i this defines a section in $\mathscr{I}^{k-1}(U_i)$.

We have that $d\beta = \alpha$ since $\alpha = \sum_{i\in\mathcal{I}} f_i\alpha|_{U_i}$. Therefore $\ker\left(\Gamma(X,\mathscr{I}^k) \to \Gamma(X,\mathscr{I}^{k+1})\right) = \operatorname{im}\left(\Gamma(X,\mathscr{I}^{k-1}) \to \Gamma(X,\mathscr{I}^k)\right)$, which means

$$\forall k > 0 \quad \mathrm{H}^k(X,\mathscr{F}) = 0$$

□

Definition 11.0.3. Let M be a smooth manifold of dimension m.
The **bundle of k-forms** $\Lambda^k(\mathrm{T}^*M)$ is the smooth vector bundle defined with:

- *Total space*: $\Lambda^k(\mathrm{T}^*M) = \coprod_{p\in M} \Lambda^k(\mathrm{T}_p^*M)$.
- *Projection map*: $\pi : \Lambda^k(\mathrm{T}^*M) \to M,\ \pi(p,\omega) = p$.
- *Fiber space*: For every $p \in M$ the fiber $E_p = \pi^{-1}(\{p\}) = \Lambda^k(\mathrm{T}_p^*M)$ is a real vector space of dimension $\binom{m}{k}$.

To define the local trivialization, consider a local chart $(U, \{x^i\}_{i=1}^m)$ of M. Then the differentials $\{\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}\}_{1\le i_1<\cdots<i_k\le m}$ form a basis for each fiber with base point in U and the map $\phi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^{\binom{m}{k}}$ is given by sending a form ω to its coefficients in this base.

Notation. Let $I = (i_1, \ldots, i_k)$ be a multi-index. We will write

$$\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$$

In this way the basis of the local trivialization is $\{\mathrm{d}x^I\}_I$.

Definition 11.0.4 (Smooth k-form)**.** Let M be a smooth manifold of dimension m.
The **sheaf of k-forms** $\Omega^k(M)$ over M is the sheaf of sections of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$.
More concretely, a k-form over an open set U is a map $\omega : U \to \Lambda^k(\mathrm{T}^*M)$ such that $\pi \circ \omega = \mathbb{1}_U$.

Example 11.0.1. Let M be a smooth manifold of dimension m.
Note that $\Lambda^0(\mathrm{T}^*M) = M \times \mathbb{R}$ is the trivial vector bundle. Therefore a 0-form is a map $\omega : M \to M \times \mathbb{R}$ such that $\pi \circ \omega = \mathbb{1}_M$, i.e: ω is a smooth function.
This means that $\Omega^0(M) \cong C^\infty(M)$.

Example 11.0.2. Let M be a smooth manifold of dimension m.
Suppose that $k > m$. Since the fibers of the bundle of k-forms $\Lambda^k(\mathrm{T}^*M)$ have dimension $\binom{m}{k}$ this means that at least one index i in the multi-index I will be repeated. Therefore since $\mathrm{d}x^i \wedge \mathrm{d}x^i = 0$ we have that $\mathrm{d}x^I = 0$, i.e: $\Lambda^k(\mathrm{T}^*M) = 0$.
In particular $\Omega^k(M) = 0$ for all $k > m$.

Definition 11.0.5 (Exterior derivative)**.** Let ω be a k-form over a smooth manifold M. If

$(U, \{x^i\}_{i=1}^m)$ is a local chart then there exist functions $\alpha_I \in C^\infty(M)$ such that

$$\omega = \sum_I \alpha_I \mathrm{d}x^I$$

where $I = (i_1, \dots, i_k)$ is a multi-index and $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$.
We can define the **exterior derivative** of ω as the $k+1$-form

$$\mathrm{d}\omega = \sum_I \mathrm{d}\alpha_I \wedge \mathrm{d}x^I = \sum_I \left(\sum_{j=1}^m \frac{\partial \alpha_I}{\partial x^j} \mathrm{d}x^j \right) \wedge \mathrm{d}x^I$$

Lemma 11.0.1. Let M be a smooth manifold of dimension m.
Then for any k-form the exterior derivative verifies $\mathrm{d}^2\omega = 0$, i.e: the sequence

$$\Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \Omega^2(M) \xrightarrow{\mathrm{d}} \cdots$$

is a complex of sheaves of $C^\infty(M)$-modules.

Proof. It suffices to show that for any smooth function $f \in C^\infty(M) \cong \Omega^0(M)$ we have $\mathrm{d}^2 f = 0$. Indeed, if $\mathrm{d}^2 f = 0$ then for any *basic* form $f\mathrm{d}x^I$ we have

$$\mathrm{d}(f\mathrm{d}x^I) = \mathrm{d}(\mathrm{d}f \wedge \mathrm{d}x^I) = \mathrm{d}^2 f \wedge \mathrm{d}x^I - \mathrm{d}f \wedge \mathrm{d}^2 x^I = 0$$

Let $f \in C^\infty(M)$ be a smooth function. Its exterior derivative is

$$\mathrm{d}f = \sum_{i=1}^m \frac{\partial f}{\partial x^i} \mathrm{d}x^i$$

Differentiating again we get

$$\mathrm{d}^2 f = \sum_{j=1}^m \frac{\partial}{\partial x^j} \left(\sum_{i=1}^m \frac{\partial f}{\partial x^i} \right) \wedge \mathrm{d}x^j = \sum_{i,j=1}^m \frac{\partial^2 f}{\partial x^i \partial x^j} \mathrm{d}x^i \wedge \mathrm{d}x^j$$

By Clairaut's theorem we have that $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ and by the antisymmetry of the wedge product $\mathrm{d}x^i \wedge \mathrm{d}x^j = -\mathrm{d}x^j \wedge \mathrm{d}x^j$. Therefore all the terms cancel leaving

$$\mathrm{d}^2 f = 0$$

□

Definition 11.0.6 (de Rham complex)**.** Let M be a smooth manifold of dimension m.
The **de Rham complex** of M is the complex of sheaves of $C^\infty(M)$-modules

$$\Omega^\bullet(M) := \Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \cdots \xrightarrow{\mathrm{d}} \Omega^m(M) \xrightarrow{\mathrm{d}} 0$$

where $\Omega^k(M)$ is the sheaf of k-forms over M and d is the exterior derivative.

Lemma 11.0.2 (Poincaré)**.** Let $U \subseteq \mathbb{R}^n$ be a contractible open set and $k \geq 1$. If $\omega \in \Omega^k(U)$ is a k-form such that $\mathrm{d}\omega = 0$ then there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(U)$ such that $\mathrm{d}\eta = \omega$.

Lemma 11.0.3. The de Rham complex is exact.

Proof. For $k \geq 1$ consider the relevant part of the complex

$$\Omega^{k-1} \xrightarrow{\mathrm{d}^{k-1}} \Omega^k \xrightarrow{\mathrm{d}^k} \Omega^{k+1}$$

We need to show that for any $p \in M$ the complex

$$\left(\Omega^{k-1}\right)_p \xrightarrow{\mathrm{d}_p^{k-1}} \left(\Omega^k\right)_p \xrightarrow{\mathrm{d}_p^k} \left(\Omega^{k+1}\right)_p$$

is exact, i.e: $\operatorname{im} \mathrm{d}_p^{k-1} = \ker \mathrm{d}_p^k$.

For the $[\subseteq]$ inclusion, let $\mathrm{d}\eta \in \operatorname{im} \mathrm{d}^{k-1} \subseteq \Omega^k$. By Lemma 11.0.1 we have that $\mathrm{d}(\mathrm{d}\eta) = 0$, i.e: $\mathrm{d}\eta \in \ker \mathrm{d}^k$.

For the $[\supseteq]$ inclusion, let $[\omega]_p \in \ker \mathrm{d}_p^k$, i.e: $[\omega]_p \in \left(\Omega^k\right)_p$ verifies $\mathrm{d}_p^k[\omega]_p = 0$. By definition there exists an open neighbourhood U of p and a k-form ω such that $\mathrm{d}^k\omega|_U = 0$.

Since M is a smooth manifold of dimension m we can choose a local chart (V, ψ) around p such that $V \subseteq U$ and $\psi(V)$ is an open ball in $\mathbb{R}^m$. Since open balls are contractible, by Poincaré's lemma there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(V)$ such that $\mathrm{d}^{k-1}\eta = \omega|_V$.

Therefore there exists $[\eta]_p \in \left(\Omega^{k-1}\right)_p$ such that $\mathrm{d}_p^{k-1}[\eta]_p = [\omega]_p$, i.e: $[\omega]_p \in \operatorname{im} d_p^{k-1}$. □

Lemma 11.0.4. Let M be a smooth paracompact manifold. Then for all k the sheaf of k-forms $\Omega^k(M)$ is fine.

Proof. Let $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ be an arbitrary open cover of M.

Since M is paracompact there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}} \subset C^\infty(M)$ subordinate to $\mathscr{U}$. This means that

$$\begin{aligned} \forall i \in \mathcal{I} \quad & \operatorname{supp} f_i \subseteq U_i \\ \forall p \in M \quad & \sum_{i\in\mathcal{I}} f_i(p) = 1 \end{aligned}$$

and this last sum is finite.

For all indices $i \in \mathcal{I}$ consider the map

$$\begin{aligned} \eta_i : \Omega^k(M) &\to \Omega^k(M) \\ \omega &\mapsto f_i \cdot \omega \end{aligned}$$

Note that since $\operatorname{supp} f_i \subseteq U_i$ we have $\operatorname{supp} \eta_i \subseteq U_i$. Furthermore for any k-form ω

$$\sum_{i\in\mathcal{I}} \eta_i(\omega) = \sum_{i\in\mathcal{I}} f_i \cdot \omega = \left(\sum_{i\in\mathcal{I}} f_i\right) \cdot \omega = 1 \cdot \omega = \omega$$

Therefore $\sum_{i\in\mathcal{I}} \eta_i = \mathbb{1}$, making $\{\eta_i\}_{i\in\mathcal{I}}$ a partition of unity subordinate to $\mathscr{U}$. □

Theorem 11.0.1. Let M be a smooth paracompact manifold. Then

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) \cong \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

Proof. Consider the resolution

$$0 \longrightarrow \underline{\mathbb{R}} \longrightarrow \Omega^\bullet(M)$$

where $\Omega^\bullet$ is the de Rham complex of M.

By Lemma 11.0.3 we know that $\Omega^\bullet(M)$ is exact, therefore we only need to show that this resolution is exact at $\Omega^0(M)$, i.e: that for all $p \in M$

$$0 \longrightarrow \underline{\mathbb{R}}_p \xrightarrow{\iota_p} \Omega^0_p \xrightarrow{\mathrm{d}_p} \Omega^1_p \longrightarrow 0$$

is a short exact sequence, i.e: $\operatorname{im} \iota_p = \ker \mathrm{d}_p$.

- $\subseteq$ Let $[c]_p \in \underline{\mathbb{R}}_p$. By definition this is a locally constant function $c : U \to \mathbb{R}$ on some open neighbourhood U of p.
  Since c is locally constant we have that $\mathrm{d}c|_U = 0$, i.e $\mathrm{d}_p[c]_p = 0$.
- $\supseteq$ Let $[f]_p \in \Omega^0_p$ such that $\mathrm{d}_p[f]_p = 0$. This means that there exists an open neighbourhood U of p and a function $f \in \Omega^0(V) \cong C^\infty(M)$ such that $\mathrm{d}f|_U = 0$. Therefore f is locally constant in each of the connected components of U, i.e: $[f]_p \in \underline{\mathbb{R}}_p$.

Therefore since by Lemma 11.0.4 the resolution is made of fine sheaves and fine sheaves are $\Gamma(X, -)$ acyclic by definition of sheaf cohomology

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) = \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

□

Corollary 11.0.1. Let M be a smooth paracompact manifold of dimension m. Then for all $k > m$

$$\mathrm{H}^k(M, \underline{\mathbb{R}}) = 0$$

Example 11.0.3 (The cohomology of S^1)**.** Consider the circle S^1 as a smooth manifold of dimension 1.
We are going to compute its sheaf cohomology with coefficients in $\underline{\mathbb{R}}$, $\mathrm{H}^\bullet(S^1, \underline{\mathbb{R}})$, using the de Rham complex

$$\Omega^0(S^1) \xrightarrow{\mathrm{d}^0} \Omega^1(S^1) \xrightarrow{\mathrm{d}^1} 0$$

On the one hand

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \ker d^0 = \{f \in \Omega^0(S^1) \cong C^\infty(S^1) \mid \mathrm{d}f = 0\}$$

Since S^1 is connected, the only functions with 0 derivative are the constant functions. Therefore

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \mathbb{R}$$

On the other

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = \ker d^1 \,/\, \operatorname{im} d^0 = \{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} \,/\, \{\mathrm{d}f \mid f \in \Omega^0(S^1) \cong C^\infty(S^1)\}$$

Note that since $\Omega^2(M) = 0$ all 1-forms $\omega \in \Omega^1(S^1)$ verify $\mathrm{d}\omega = 0$. Therefore

$$\{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} = \Omega^1(S^1)$$

Consider now the *integration* map

$$I : \Omega^1(S^1) \to \mathbb{R}$$
$$\omega \mapsto \int_{S^1} \omega$$

Let $\mathrm{d}\theta$ be the differential of the angle. Since $I(\mathrm{d}\theta) = 2\pi$ we have that I is surjective. Furthermore by Stoke's theorem we have that $I(\omega) = 0$ if and only if $w = \mathrm{d}f$ for some $f \in C^\infty(S^1)$.
Therefore by the first isomorphism theorem we have that

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = {}^{\Omega^1(S^1)}\big/_{\ker I} \cong \operatorname{im} I = \mathbb{R}$$

In summary, we have that

$$\mathrm{H}^k(S^1, \underline{\mathbb{R}}) = \begin{cases} \mathbb{R} & k \in \{0, 1\} \\ 0 & k > 1 \end{cases}$$

Appendix A

Algebra

Definition A.0.1 (Algebra)**.**

Definition A.0.2 (Local ring)**.**

Definition A.0.3 (Module)**.**

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